Eigenvector Occupation Statistics and Extremes in the Rosenzweig–Porter Model

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14.01.2077

Abstract

We study the occupation probabilities $p_i = |u_i|^2$ of a normalized bulk eigenvector in the GUE Rosenzweig–Porter model $A_N = B_N + \tau_N G_N$, with $\tau_N^2 = N^{1-\gamma}$ and $1 < \gamma < 2$. Conditioning on an eigenvalue at a regular bulk energy E through the normalized one-eigenpair Kac–Rice law, we derive an exact finite-N projective density and reduce its asymptotic analysis to a Hubbard–Stratonovich integral over 2×2 Hermitian matrices, which we evaluate by the saddle-point method. The resulting occupation field is governed by a self-consistent Lorentzian profile with effective resonant size $K_N(E) \asymp N^{2-\gamma}$ and local exponential fluctuations. We prove the asymptotics of all generalized inverse participation sums and identify a transition at $q = 1/2$. We also obtain a two-branch thickened amplitude-counting spectrum. At the two ends of the occupation field we prove marked Poisson point-process limits: the largest components have Gumbel fluctuations, whereas the smallest component has an exponential ...limit on the scale $N^{-1-\gamma}$. ~~Finally,~~ Furthermore, for every fixed $q > 0$ the largest component... contributes an asymptotically vanishing fraction of the inverse participation sum, showing that these fixed-q statistics do not freeze. The statements are quenched, uniformly on a high-probability regularity event for the diagonal disorder, and hence also hold in the annealed setting.

Keywords. Rosenzweig–Porter model; eigenvectors; occupation probabilities; inverse participation ratios; multifractality; Poisson point processes; extreme-value statistics; Kac–Rice formula.

Mathematics Subject Classification (2077). 60B20, 60G70, 82B44.

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1 Introduction

~~The Rosenzweig-Porter (RP) ensemble is one of the simplest random-matrix models exhibiting non-ergodic delocalization.~~ The Rosenzweig-Porter (RP) ensemble, originally proposed to describe complex atomic spectra, serves as a paradigmatic and analytically tractable random-matrix model exhibiting a robust non-ergodic delocalized (NED) phase.: its norm is spread over $O(N)$ coordinates, with typical occupation weights $p_i = O(N^{-1})$. At the opposite extreme, in the localized regime of a random Schrödinger operator, an eigenfunction has appreciable weight on only $O(1)$ sites. Critical wave functions at an Anderson transition provide an intermediate possibility, in which the number of significantly occupied sites diverges with the system size but remains sublinear in the total volume [**?**, **?**]. The RP ensemble realizes an analogous intermediate regime in a particularly tractable setting, without an underlying spatial geometry. In its non-ergodic delocalized phase, an eigenvector occupies a diverging but vanishing fraction of the N available coordinates. The model therefore isolates the basic resonance mechanism responsible for such states while remaining simple enough to permit a detailed analysis of the full occupation statistics.

We consider

$$A_N = B_N + \tau_N G_N, \qquad B_N = \mathrm{diag}(b_1, \dots, b_N), \qquad \tau_N^2 = N^{1-\gamma}, \qquad 1 < \gamma < 2,$$

where G_N is GUE and the b_i are independent samples from a regular density ρ. If u is a normalized eigenvector, its occupation weights

$$p_i = |u_i|^2, \qquad \sum_{i=1}^{N} p_i = 1,$$

form a random probability measure on the diagonal sites. The generalized inverse participation sums

$$I_q = \sum_{i=1}^{N} p_i^q$$

record moments of this measure. ~~In terminology customary in physics,~~ In the standard terminology used in physics, the distribution of components over the polynomial scales $p_i \geq N^{-\alpha}$ is encoded in the multifractal, or singularity, spectrum $f(\alpha)$. Through "thickened" amplitude counts: one counts the components in the logarithmic window

$$N^{-\alpha-\varepsilon} \leq p_i \leq N^{-\alpha+\varepsilon}$$

and lets $\varepsilon \downarrow 0$ after taking $N \to \infty$.

The purpose of this paper is to describe all three aspects at a fixed regular bulk energy: the generalized inverse participation sums, the multifractal spectrum $f(\alpha)$, and the upper and lower extreme-value statistics of the occupation weights, including the largest and smallest weights

$$p_{\max} = \max_i p_i, \qquad p_{\min} = \min_i p_i.$$

1.1 Background and motivation

Rosenzweig and Porter introduced their ensemble as an interpolation between Poisson and Wigner–Dyson level statistics in complex atomic spectra [**?**]. The spectral crossover in the RP ensemble was subsequently studied by several complementary random-matrix methods [**?**, **?**]. Closely related exact formulas for spectral correlations in the external-source ensemble, consisting of a deterministic Hermitian matrix perturbed by a GUE matrix, were derived in [**?**]. Conditioned on its diagonal part, the RP ensemble has precisely this deformed-GUE structure, although with an N-dependent perturbation strength. In the generalized RP scaling, where the strength of the Gaussian perturbation depends on N, the phase diagram comprises an ergodic phase, a localized phase, and an intermediate non-ergodic delocalized phase. This three-phase structure was first identified systematically by Kravtsov, Khaymovich, Cuevas et al. in [**?**]. In the convention of the present paper, the intermediate phase is $1 < \gamma < 2$ and the characteristic number of resonant coordinates is of order

$$N^{2-\gamma}.$$

Thus a bulk eigenvector is extended over a diverging number of sites but over a vanishing fraction of the ambient space.

The physics literature has developed a detailed description of RP eigenfunctions using resolvent, perturbative, replica, supersymmetric, and dynamical methods. In particular, the Lorentzian resonant profile and the participation exponents were obtained in [**?**, **?**, **?**], while Bogomolny and Sieber derived an explicit one-component eigenfunction distribution from the Breit–Wigner profile and a local Porter–Thomas ansatz [**?**]. Replica and dynamical aspects were further studied in [**?**, **?**, **?**], and the local miniband viewpoint was developed in [**?**].

It is useful to distinguish clearly between those of our results that rigorously confirm this physical picture and those that provide information beyond it. The scaling exponents of the inverse participation sums, including their change of behaviour at $q = 1/2$, and the conventional RP branch

$$f(\alpha) = 1 - \frac{\gamma}{2} + \frac{\alpha}{2}, \qquad 2 - \gamma \le \alpha \le \gamma,$$

were already predicted in the physics literature. Moreover, for the Gaussian RP ensemble at the centre of the spectrum, Bogomolny and Sieber obtained explicit leading prefactors and certain finite-N corrections to the disorder-averaged moments [**?**]. We therefore do not claim a more precise asymptotic expansion of the annealed inverse participation sums. Our result is different in nature: it establishes the corresponding quenched statement. For a typical fixed realization of the diagonal disorder, the actual inverse participation sums concentrate around their explicitly determined leading values, uniformly over regular bulk energies.

A one-component distribution, however, does not determine the joint rare-event statistics of all N occupation weights. In particular, although a Gumbel law for the maximum may be suggested by a heuristic picture of approximately independent tails, it does not follow from the one-component distribution alone. Correlations between the components, their site-dependent scales, and the common normalization $\sum_i p_i = 1$ may all affect the extreme-value statistics. Our results establish the Poisson convergence of the upper extreme process and thereby prove the Gumbel law for the maximum, including its precise centering. They also establish the Poisson limit at the lower edge, the resulting law of the minimum, and, jointly with the extreme weights, determine the distribution of the diagonal energies of the sites on which these extreme weights occur. This global and extreme-value information is the principal new content of the present paper.

Several rigorous results establish important parts of the deformed-Wigner and RP picture. Von Soosten and Warzel proved non-ergodic delocalization on the expected resonant scale [**?**]. Benigni obtained asymptotic Gaussianity and a deformed quantum unique ergodicity statement for mesoscopic deformed Wigner matrices [**?**]. Related perturbative spectral-measure results were proved by Benaych-Georges, Enriquez, and Michaïl [**?**], while Mészáros and Virág analysed Gaussian-wave limits for a concrete deformed model [**?**]. Very recent work of Cipolloni, Erdős, and Henheik treats local laws, thermalization, and localization questions for a broad RP-type class [**?**]. The present paper complements this literature by obtaining occupation observables and extreme-value point processes directly from a one-eigenpair law.

At the level of model building, the RP ensemble stands in relation to multifractal random Schrödinger operators much as Derrida's Random Energy Model (REM) stands in relation to logarithmically correlated random-energy landscapes. The REM provides a solvable independent-energy backbone which retains the competition between the entropy of the available states and the statistical advantage of rare energies [**?**]. Logarithmic REMs enrich this picture by restoring correlations across scales, thereby producing a more structured multifractal and extreme-value phenomenology [**?**, **?**]. The same log-correlated freezing paradigm also appears in the extreme-value statistics of random-matrix characteristic polynomials and has been conjecturally extended to the Riemann zeta function [**?**].

In the same way, the RP ensemble provides a geometry-free, all-to-all backbone for resonance-driven eigenfunction statistics. It retains the competition between the number of available sites and the enhancement produced by near resonances, but suppresses the spatial organization and correlations present in genuine random Schrödinger operators. From this perspective, spatially correlated extensions of the RP mechanism play a role analogous to logarithmically correlated extensions of the REM: they provide a bridge from the solvable mean-field model towards the richer multifractality of spatially disordered systems. On the Schrödinger side, a recent rigorous result by Keating and Ueberschär proves, in dimensions $d \geq 5$, an explicit nonconstant spectrum of Rényi exponents for an averaged Fourier-space measure associated with eigenfunctions of Schrödinger operators with point-like scatterers in the thermodynamic limit, under suitable weak-disorder and strong-coupling assumptions [**?**]. Complementary routes towards richer, genuinely multifractal regimes have been investigated in RP-inspired ensembles with structured disorder and correlated or spatially decaying hopping [**?**, **?**].

1.2 Main results and method

Our starting point is the normalized Kac–Rice Palm law of an eigenvector conditioned on the occurrence of an eigenvalue at energy E. For a fixed diagonal realization B_N, we derive an exact finite-N density on complex projective space. A fermionic representation of the tangent determinant and a Hubbard–Stratonovich transform reduce its asymptotic analysis to an integral over 2×2 Hermitian matrices. The idea of using the Kac–Rice counting formula to derive joint eigenvalue–eigenvector densities for random matrices was developed in [**?**], initially in the non-Hermitian setting. Here we adapt this construction to the Hermitian Rosenzweig–Porter ensemble and to the projective law of an eigenvector conditioned on an eigenvalue at the fixed energy E. The subsequent fermionic and Hubbard–Stratonovich reduction is closely related to the supersymmetric method [**?**]; the mixed-sign saddle analysis is also closely related to characteristic-polynomial asymptotics in [**?**, **?**, **?**].

The self-consistent saddle produces deterministic weights

$$m_i(E) = \frac{\tau_N^2}{N((b_i - E + t_N(E))^2 + \eta_N(E)^2)}, \qquad \sum_i m_i(E) = 1,$$

with $\eta_N(E) \sim \pi\rho(E)\tau_N^2$. Their effective resonant number is

$$K_N(E) = \frac{N\eta_N(E)^2}{\tau_N^2} \sim \pi^2\rho(E)^2 N^{2-\gamma}.$$

At the observable level, the occupation weights behave as the Lorentzian profile multiplied by local mean-one exponential variables. The exact normalization constraint is retained throughout the Kac–Rice calculation.

The consequences are as follows.

1. For every fixed $q > 0$, the generalized inverse participation sum $I_q = \sum_i p_i^q$ is asymptotic to $\Gamma(q+1)\sum_i m_i(E)^q$. The deterministic sum changes mechanism at $q = 1/2$: resonant coordinates dominate for $q > 1/2$, while the bulk of off-resonant coordinates contributes for $q < 1/2$; the critical case has a logarithmic correction.

2. The number of coordinates with $p_i \asymp N^{-\alpha}$ has a two-branch exponent $f(\alpha)$. The increasing branch comes from the Lorentzian spatial profile, whereas the decreasing branch is generated by the small-value tail of the local exponential factor. This gives a degenerate piecewise-linear form of multifractality, rather than merely a single support dimension.

3. The upper and lower extremes have different mechanisms. After rescaling, the largest occupations form a marked Poisson process and their maximum has a Gumbel limit, with a $-\frac{1}{2}\log\log K_N$ correction to the centering. The smallest occupations form a second marked Poisson process and $N^{1+\gamma}\mu_2(E)p_{\min}$ converges to a mean-one exponential law. The associated diagonal marks distinguish the resonant origin of the upper extremes from the off-resonant origin of the lower ones.

Despite the nontrivial upper extremes, there is no condensation at fixed q: we prove $p_{\max}^q/I_q \to 0$ in probability for every fixed $q > 0$. Thus the Gumbel maximum does not produce logREM-type freezing of the physical occupation vector.

All asymptotic statements are first proved quenched, uniformly on a high-probability regularity event for B_N, and therefore pass to the annealed law. Section 2 defines the model, the Kac–Rice Palm law, and states the results. Section 3 derives the exact one-frame density and its integral representation. Section 4 performs the saddle analysis and proves the limit theorems. The Appendix collects the uniform empirical and self-consistency estimates used in the proof.

2 Model

Let

$$A_N = B_N + \tau_N G_N,$$

where $B_N = \mathrm{diag}(b_1, \dots, b_N)$ and G_N is a GUE matrix. Assume

$$\tau_N^2 = N^{1-\gamma}, \quad 1 < \gamma < 2. \tag{2.1}$$

The diagonal variables $b_1, \dots, b_N$ are iid with density ρ. We assume that $\rho \in C^3(\mathbb{R})$ and decays sufficiently fast at infinity to have a well-defined Hilbert transform, and $\rho(x) > 0$ on a fixed open interval containing the compact bulk interval I under consideration. We also assume that ρ has a finite second moment and denote

$$\mu_2(E) = \int (b - E)^2 \rho(b) db. \tag{2.2}$$

For $E \in I$, let $t_N(E)$ and $\eta_N(E) > 0$ be the solution of

$$t_N = \frac{\tau_N^2}{N} \sum_{i=1}^{N} \frac{E - t_N - b_i}{(b_i - E + t_N)^2 + \eta_N^2}, \qquad 1 = \frac{\tau_N^2}{N} \sum_{i=1}^{N} \frac{1}{(b_i - E + t_N)^2 + \eta_N^2}. \tag{2.3}$$

As we will prove below, such a solution exists and is unique in the region $|t| \le C\tau_N^2$, $c\tau_N^2 \le \eta \le C\tau_N^2$, which is the regime used below.

Put

$$m_i(E) = \frac{\tau_N^2}{N[(b_i - E + t_N(E))^2 + \eta_N(E)^2]}, \qquad \sum_{i=1}^{N} m_i(E) = 1, \qquad K_N(E) = \frac{N\eta_N(E)^2}{\tau_N^2}. \tag{2.4}$$

We will also prove

$$\eta_N(E) = \pi\rho(E)\tau_N^2[1 + o(1)], \qquad K_N(E) = \pi^2\rho(E)^2 N\tau_N^2[1 + o(1)] \longrightarrow \infty. \tag{2.5}$$

2.1 Main results

Let $B_N = \mathrm{diag}(b_1, \dots, b_N)$ be fixed. Let also $d\omega_{FS}$ be the normalized Fubini–Study measure on $\mathbb{C}P^{N-1} = \{u \in \mathbb{C}^N : \|u\| = 1\}/\{u \sim e^{i\theta}u\}$: it is the push-forward of normalized Haar measure $d\sigma$ on the unit sphere $S^{2N-1} \subset \mathbb{C}^N$ under $u \mapsto [u]$, where $[u]$ is the eigenvector line with the irrelevant complex phase quotiented out. Define the one-eigenpair intensity by

$$\mathbb{E} \sum_{\alpha=1}^{N} F(\lambda_\alpha, [\psi_\alpha]) = \int F(E, [u]) \rho_{B_N}^{(1)}(E, [u])\, dE\, d\omega_{FS}([u]),$$

and the Kac-Rice law

$$d\mu_{N,E}^{KR}([u]) = \frac{\rho_{B_N}^{(1)}(E, [u]) d\omega_{FS}([u])}{\int_{\mathbb{C}P^{N-1}} \rho_{B_N}^{(1)}(E, [u]) d\omega_{FS}([u])}. \tag{2.6}$$

Let u be sampled from this law and

$$p_i = |u_i|^2. \tag{2.7}$$

Let Ω_N^{loc} be the event in the probability space of B_N on which Proposition A.1 and Lemma A.2 hold. Its complement has probability at most N^{-D} for every fixed $D > 0$. Choose a deterministic sequence $\delta_N \downarrow 0$ such that

$$\mathbb{P}\left\{ \sup_{E \in I} \left| \frac{1}{N} \sum_{i=1}^{N} (b_i - E)^2 - \mu_2(E) \right| > \delta_N \right\} \longrightarrow 0, \tag{2.8}$$

$$\mathbb{P}\left\{ \sup_{E \in I} \sup_{\|F\|_\infty + \mathrm{Lip}(F) \le 1} \left| \frac{1}{N} \sum_{i=1}^{N} (b_i - E)^2 F(b_i) - \int (b - E)^2 F(b) \rho(b)\, db \right| > \delta_N \right\} \longrightarrow 0,$$

and, after making δ_N decrease more slowly if necessary, also

$$\mathbb{P}\left\{\frac{1}{N}\max_{1\le i\le N} b_i^2 > \delta_N\right\} \longrightarrow 0. \tag{2.9}$$

Let $\Omega_N^{(2)}$ be the intersection of the three corresponding good events. Such a choice is possible by the strong law of large numbers, compactness of I, a standard truncation argument for the weighted empirical measures, and the consequence of $\mathbb{E}b_1^2 < \infty$ that $N^{-1}\max_i b_i^2 \to 0$ in probability. Finally, set

$$\Omega_N := \Omega_N^{\mathrm{loc}} \cap \Omega_N^{(2)}.$$

Thus $\mathbb{P}(\Omega_N^c) \to 0$.

Then we have the following theorems

Theorem 2.1 (Inverse participation sums)**.** *Fix $q > 0$ and set*

$$I_q = \sum p_j^q \tag{2.10}$$

On the event Ω_N, under the quenched Kac–Rice law, we have

$$\frac{I_q}{\Gamma(q+1)\sum_{i=1}^N m_i(E)^q} \longrightarrow 1 \qquad \textit{in probability.} \tag{2.11}$$

The convergence is uniform in $B_N \in \Omega_N$ and therefore also holds under the annealed law. The sum in the denominator has the following asymptotics:

$$\sum_i m_i^q = K_N^{1-q}\frac{\Gamma(q-1/2)}{\sqrt{\pi}\,\Gamma(q)}[1+o_{\mathbb{P}}(1)], \qquad q > \frac{1}{2}, \tag{2.12}$$

$$\sum_i m_i^{1/2} = \frac{2}{\pi}K_N^{1/2}\log\frac{1}{\eta_N}[1+o_{\mathbb{P}}(1)], \qquad q = \frac{1}{2}, \tag{2.13}$$

$$\sum_i m_i^q = N^{1-\gamma q}\left[\int_{\mathbb{R}} |x-E|^{-2q}\rho(x)\,dx + o_{\mathbb{P}}(1)\right], \qquad 0 < q < \frac{1}{2}. \tag{2.14}$$

All asymptotic relations here are in probability.

For $\alpha \in \mathbb{R}$ and $\varepsilon > 0$, let

$$\mathcal{N}_N(\alpha,\varepsilon) = \#\{i : N^{-\alpha-\varepsilon} \le p_i \le N^{-\alpha+\varepsilon}\}.$$

Theorem 2.2 (Thickened amplitude-counting spectrum)**.** *On Ω_N, for every α in the interior of a region where the function below is positive, under the quenched Kac–Rice law, we have*

$$\lim_{\varepsilon\downarrow 0}\lim_{N\to\infty}\frac{\log\mathcal{N}_N(\alpha,\varepsilon)}{\log N} = f(\alpha) \quad \textit{in probability,}$$

where

$$f(\alpha) = \begin{cases} 1-\gamma/2+\alpha/2, & 2-\gamma \le \alpha \le \gamma, \\ 1+\gamma-\alpha, & \gamma < \alpha \le \gamma+1. \end{cases}$$

The convergence is uniform in $B_N \in \Omega_N$ and therefore also holds under the annealed law.

Physical origin of the two branches. The two branches in Theorem 2.2 arise from two different mechanisms. For

$$2-\gamma \le \alpha \le \gamma,$$

the scale $m_i(E) \asymp N^{-\alpha}$ is produced by diagonal energies at distance

$$|b_i - E| \asymp N^{-(\gamma-\alpha)/2}$$

from the observation energy. The expected number of such sites is therefore of order

$$N\, N^{-(\gamma-\alpha)/2} = N^{\,1-\gamma/2+\alpha/2}.$$

This gives the left, blue branch

$$f_{\mathrm{L}}(\alpha) = 1 - \frac{\gamma}{2} + \frac{\alpha}{2},$$

which is the conventional RP multifractal spectrum already discussed in the physics literature.

The right, orange branch has a different origin. At a generic off-resonant site one has $m_i(E) \asymp N^{-\gamma}$, while the local GUE Porter–Thomas factor is exponential:

$$p_i \simeq m_i(E)\xi_i,$$

where $\xi_i \sim \mathrm{Exp}(1)$, i.e. ξ_i has the probability density $\mathcal{P}(\xi)\,d\xi = e^{-\xi}\,d\xi, \qquad \xi \ge 0.$

To obtain a smaller occupation $p_i \asymp N^{-\alpha}$ with $\alpha > \gamma$, one needs

$$\xi_i \asymp N^{-(\alpha-\gamma)}.$$

Since the exponential density is nonzero at the origin, among N sites the number of such small factors is of order

$$N\, N^{-(\alpha-\gamma)} = N^{\,1+\gamma-\alpha}.$$

This gives

$$f_{\mathrm{R}}(\alpha) = 1 + \gamma - \alpha, \qquad \gamma < \alpha \le \gamma + 1.$$

The endpoint $\alpha = \gamma + 1$ is where the expected number of such components becomes of order one.

The small-Porter–Thomas mechanism is implicit in the one-component distribution of Bogomolny and Sieber [**?**]. To the best of our knowledge, however, the resulting right branch was not previously isolated as part of the RP amplitude-counting spectrum. The new statement here is its explicit identification and theorem-level proof for the GUE model, together with the subsequent Poisson limits for the upper and lower extreme processes. Figure 1 displays the two branches for $\gamma = 3/2$.

Let α_N be the unique solution, for large N, of

$$\frac{K_N(E)e^{-\alpha_N}}{\sqrt{\pi\alpha_N}} = 1. \tag{2.15}$$

Equivalently,

$$\alpha_N = \log K_N(E) - \frac{1}{2}\log\log K_N(E) - \frac{1}{2}\log\pi + o(1). \tag{2.16}$$

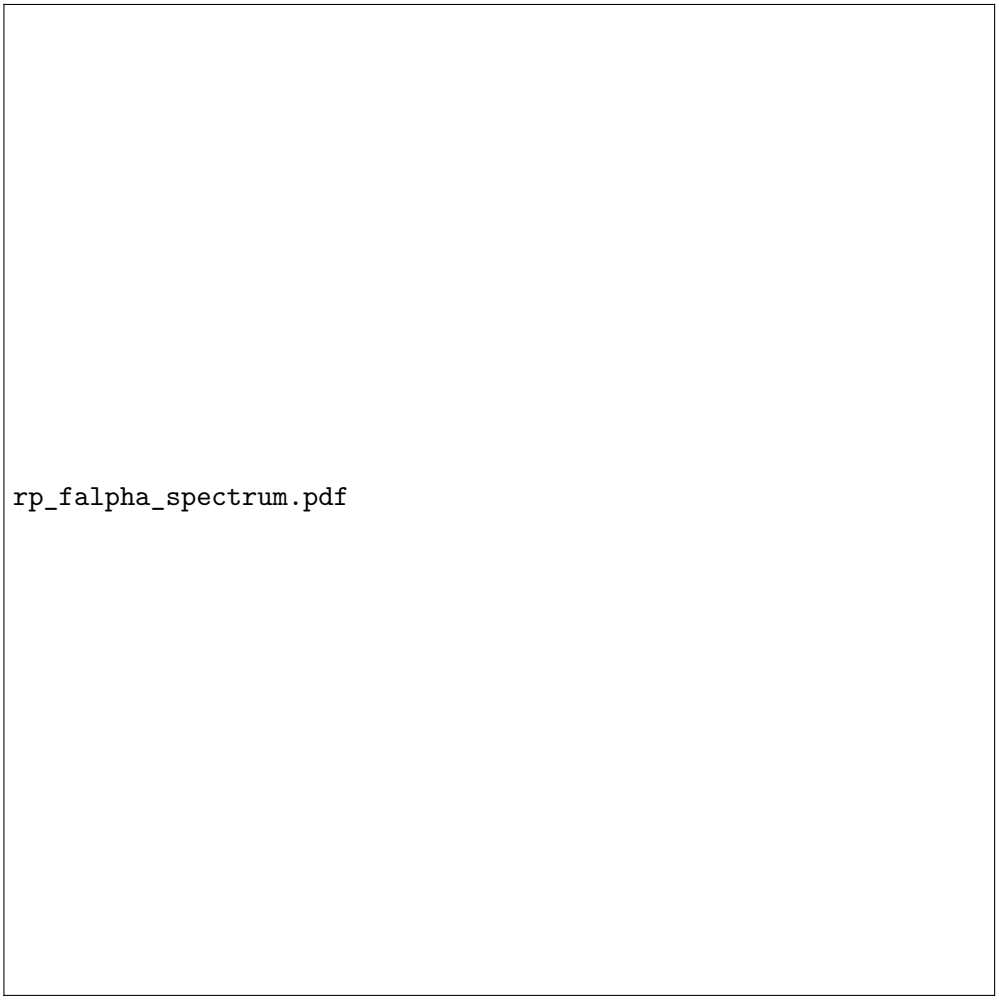


Figure 1: The thickened amplitude-counting spectrum $f(\alpha)$ for $\gamma = 3/2$, drawn at $E = 0$. The left branch is the Lorentzian structural spectrum, and the right branch is the small-amplitude exponential tail. The same exponents hold at every regular bulk energy covered by Theorem 2.2; the energy dependence enters only through the non-universal prefactors.

Theorem 2.3. *Define*

$$\beta_i = b_i - E + t_N, \qquad v_i = \sqrt{\alpha_N}\,\frac{\beta_i}{\eta_N}, \tag{2.17}$$

and the point process

$$\Xi_N^+ = \sum_{i=1}^{N} \delta_{(v_i,\, K_N p_i - \alpha_N)}.$$

Then, under the quenched Kac–Rice law, uniformly in $B_N \in \Omega_N$ and $E \in I$,

$$\Xi_N^+ \Longrightarrow \mathrm{PPP}\left(\frac{e^{-v^2}}{\sqrt{\pi}}\,dv\,e^{-x}\,dx\right),$$

vaguely on $\mathbb{R} \times (-A, \infty)$, for every fixed A. The same conclusion holds under the annealed law.

Corollary 2.1. *Let*

$$p_{max} = \max_{1\le i\le N} p_i.$$

Then on Ω_N, under the same quenched and annealed laws,

$$p_{max} = \frac{1}{K_N(E)}(\log K_N(E) - \frac{1}{2}\log\log K_N(E) - \frac{1}{2}\log\pi + O_{\mathbb{P}}(1)).$$

More precisely,

$$K_N p_{\max} - \alpha_N \Longrightarrow \mathcal{G},$$

where $\mathcal{G}$ has the Gumbel distribution

$$\mathbb{P}\{\mathcal{G} \le x\} = e^{-e^{-x}}.$$

The convergence is uniform in $B_N \in \Omega_N$ and therefore also holds under the annealed law.

Corollary 2.2 (Absence of freezing at fixed q)**.** *For every fixed $q > 0$, under the same quenched and annealed laws,*

$$\frac{p_{\max}^q}{I_q} \xrightarrow{\mathbb{P}} 0. \tag{2.18}$$

Thus, although $p_{\max}$ has nontrivial Gumbel fluctuations on its natural extreme-value scale, the largest component carries an asymptotically vanishing fraction of every fixed positive inverse participation sum I_q. In this sense the fixed-q statistics remain collective and do not freeze onto the largest component.

Proof. Corollary 2.1 gives

$$p_{\max} = O_{\mathbb{P}}\left(\frac{\log K_N}{K_N}\right).$$

If $q > 1/2$, Theorem 2.1 yields

$$I_q = C_q K_N^{1-q}[1 + o_{\mathbb{P}}(1)], \qquad C_q = \frac{\Gamma(q+1)\Gamma(q-1/2)}{\sqrt{\pi}\,\Gamma(q)} > 0,$$

and hence

$$\frac{p_{\max}^q}{I_q} = O_{\mathbb{P}}\left(\frac{(\log K_N)^q}{K_N}\right) = o_{\mathbb{P}}(1).$$

For $q = 1/2$, the critical asymptotics in Theorem 2.1 give

$$\frac{p_{\max}^{1/2}}{I_{1/2}} = O_{\mathbb{P}}\left(\frac{(\log K_N)^{1/2}}{K_N \log(1/\eta_N)}\right) = o_{\mathbb{P}}(1).$$

Finally, if $0 < q < 1/2$, then $K_N \asymp N^{2-\gamma}$ and

$$I_q = C_q(E) N^{1-\gamma q}[1 + o_{\mathbb{P}}(1)], \qquad C_q(E) = \Gamma(q+1)\int_{\mathbb{R}} |x - E|^{-2q}\rho(x)\,dx > 0.$$

Therefore

$$\frac{p_{\max}^q}{I_q} = O_{\mathbb{P}}\left((\log N)^q N^{-1+2q(\gamma-1)}\right) = o_{\mathbb{P}}(1),$$

because $2q(\gamma - 1) < 1$. □

Theorem 2.4. *Define*

$$\mathcal{A}_N(E) := c_N N \mu_2(E), \tag{2.19}$$

with $\mu_2(E)$ defined in (2.2), and the point process

$$\mathcal{L}_N^- := \sum_{i=1}^N \delta_{(\mathcal{A}_N(E)p_i, b_i)}.$$

Then, under the quenched Kac–Rice law, uniformly in $B_N \in \Omega_N$ and $E \in I$,

$$\mathcal{L}_N^- \Longrightarrow \mathrm{PPP}\big(dx\, \nu_E(db)\big) \quad \textit{vaguely on } (0,\infty)\times\mathbb{R},$$

where

$$\nu_E(db) = \frac{(b-E)^2\rho(b)}{\mu_2(E)}\, db. \tag{2.20}$$

The same conclusion holds under the annealed law.

Corollary 2.3. *Let $p_{\min} = \min_i p_i$. Under the same quenched and annealed laws,*

$$\mathcal{A}_N(E) p_{\min} \Longrightarrow \mathrm{Exp}(1),$$

or, equivalently,

$$N^{1+\gamma}\mu_2(E) p_{\min} \Longrightarrow \mathrm{Exp}(1).$$

3 Exact one-frame Kac–Rice density

3.1 Kac-Rice formula

Let u be the eigenvector corresponding to the eigenvalue E. Set $P_u = I - u \otimes u^*$ and let $V_u : \mathbb{C}^{N-1} \to u^\perp$ be an isometry. Then for the Kac-Rice law (2.6) we have

Proposition 3.1. *If K is an independent $(N-1)\times(N-1)$ GUE matrix with $d\mathbb{P}(K) \propto e^{-N\mathrm{Tr}\, K^2/2} dK$, then*

$$\begin{aligned}\rho^{(1)}_{B_N}(E,[u]) = C_{N,\tau_N} \exp\left\{-\frac{N}{\tau_N^2}\|P_u B_N u\|^2 - \frac{N}{2\tau_N^2}(E - u^* B_N u)^2\right\} \\ \times \mathbb{E}_K \left|\det(\tau_N K + V_u^*(B_N - E)V_u)\right|^2.\end{aligned} \tag{3.1}$$

Moreover, denote $p_i = |u_i|^2$, $\tilde{b}_i = b_i - E$, and

$$\Delta_{N-1}(p) = \{p_i \geq 0,\, i = 1,..,N, \sum_{i=1}^N p_i = 1\}.$$

Fixing E, (3.1) can be rewritten as

$$d\mu^{KR}_{N,E}(p) \propto \mathbf{1}_{\Delta_{N-1}}(p) e^{-\frac{N}{\tau_N^2} Q_E(p)} D_{N,E}(p)\, dp, \tag{3.2}$$

where dp is Lebesgue measure,and

$$Q_E(p) = \sum_i \tilde{b}_i^2 p_i - \frac{1}{2}\left(\sum_i \tilde{b}_i p_i\right)^2, \qquad D_{N,E}(p) = \mathbb{E}_K\left|\det(\tau_N K + V_u^*(B_N - E)V_u)\right|^2. \tag{3.3}$$

Proof. Fix u with $u^*u = 1$, and choose V_u as above. Set $U = (u, V_u) \in U(N)$. The key ingredient of this proposition is the projective Kac–Rice eigenpair formula, which is the Hermitian specialization of the eigenvalue–eigenvector density construction developed in [**?**] for several classes of non-Hermitian random matrices. Up to a normalization constant depending only on the chosen conventions for $d\omega_{FS}$ and the delta functions, the formula implies

$$\rho^{(1)}_{B_N}(E, [u]) \propto \mathbb{E}_{G_N}\left[\delta_{\mathbb{R}}(u^*(A_N - E)u)\delta_{\mathbb{C}^{N-1}}(V_u^*(A_N - E)u)\left|\det(V_u^*(A_N - E)V_u)\right|^2\right].$$

By GUE invariance,

$$U^*G_NU = \begin{pmatrix} h & \xi^* \\ \xi & K \end{pmatrix},$$

where $h \in \mathbb{R}$, $\xi \in \mathbb{C}^{N-1}$, and $K \in \mathrm{Herm}(N-1)$ are independent, and the scalar h has density $e^{-Nh^2/2}$, the complex vector ξ has density $e^{-N\|\xi\|^2}$, and K is distributed as an $(N-1)\times(N-1)$ GUE matrix with $d\mathbb{P}(K) \propto e^{-N\mathrm{Tr}\,K^2/2}dK$.

In the same basis,

$$U^*(A_N - E)U = \begin{pmatrix} \tau_N h + u^*Bu - E & \tau_N\xi^* + u^*BV_u \\ \tau_N\xi + V_u^*Bu & \tau_N K + V_u^*(B - E)V_u \end{pmatrix}.$$

The eigenpair equations from the delta-functions force

$$\tau_N h + u^*B_Nu - E = 0 \Longrightarrow h = \tau_N^{-1}(E - u^*B_Nu),$$
$$\tau_N\xi + V_u^*B_Nu = 0 \Longrightarrow \xi = -\tau_N^{-1}V_u^*B_Nu.$$

Since $VV^* = P_u$, the vector term can be written as

$$\|V^*Bu\|^2 = \|P_uBu\|^2.$$

This gives the exponential factor in (3.1); the free quotient block has the displayed determinant-square tangent Jacobian.

To prove (3.2), write

$$u^*B_Nu = \sum_i b_ip_i = E + \sum_i \tilde{b}_ip_i, \qquad \|P_uB_Nu\|^2 = \sum_i b_i^2p_i - \left(\sum_i b_ip_i\right)^2.$$

Hence, we obtain

$$\|P_uB_Nu\|^2 + \frac{1}{2}(E - u^*B_Nu)^2 = \sum_i \tilde{b}_i^2p_i - \frac{1}{2}\left(\sum_i \tilde{b}_ip_i\right)^2 = Q_E(p).$$

Finally, changing the phases of u conjugates the compression $V_u^*(B_N - E)V_u$ by a unitary operator. The GUE block K is unitarily invariant, so the determinant expectation depends only on p; denote it by $D_{N,E}(p)$. Substitution in (3.1) gives (3.2). □

3.2 Integral representation for $D_{N,E}(p)$

Proposition 3.2. *The determinant factor $D_{N,E}(p)$ from (3.3) can be represented as*

$$D_{N,E}(p) = \mathcal{C}_{N,\tau_N}\int_{\mathrm{Herm}(2)} e^{-\frac{N}{2\tau_N^2}\mathrm{Tr}\,Y^2}\prod_i \det(\tilde{b}_iI_2 + iY)\det\left[\sum_i p_i(\tilde{b}_iI_2 + iY)^{-1}\right]dY. \tag{3.4}$$

Proof. Let $d = N - 1$, and put

$$C = V_u^*(B_N - E)V_u.$$

For two Grassmann fields $\Psi_\alpha = \{\psi_{\ell,\alpha}\}_l, \bar\Psi_\alpha = \{\bar\psi_{\ell,\alpha}\}_l$ $(1 \le \ell \le d,\ 1 \le \alpha \le 2)$, write the standard determinant representation via Grassmann integral

$$\int \exp\Big\{\Psi^+ M \Psi\Big\} \prod_j d\bar\psi_j\, d\psi_j = \det M$$

valid for any $d \times d$ matrix M.

It gives

$$|\det(\tau_N K + C)|^2 = \int \exp\left\{\sum_{\alpha=1}^{2} \Psi_\alpha^+ (\tau_N K + C)\Psi_\alpha\right\} d\Psi, \quad d\Psi = \prod_{\alpha=1}^{2}\prod_{j=1}^{d} d\bar\psi_{j,\alpha}\, d\psi_{j,\alpha}.$$

Define the Grassmann-even $d \times d$ matrix

$$R_{jk} := \sum_{\alpha=1}^{2} \bar\psi_{j,\alpha}\psi_{k,\alpha},$$

and the 2×2 matrix

$$S_{\alpha\beta} := \sum_{j=1}^{d} \bar\psi_{j,\alpha}\psi_{j,\beta}.$$

The GUE average gives

$$\mathbb{E}_K e^{\tau_N \operatorname{Tr}(KR)} = \exp\left\{\frac{\tau_N^2}{2N} \operatorname{Tr} R^2\right\}.$$

Anticommutation of the Grassmann variables yields

$$\operatorname{Tr} R^2 = -\operatorname{Tr} S^2.$$

Consequently,

$$D_{N,E}(p) = \int \exp\left\{\sum_{\alpha=1}^{2} \Psi_\alpha^+ C \Psi_\alpha - \frac{\tau_N^2}{2N} \operatorname{Tr} S^2\right\} d\Psi.$$

Applying the Hubbard-Stratonovich transformation (see, e.g., [**?**]), we get

$$e^{-\frac{\tau_N^2}{2N}\operatorname{Tr} S^2} = C_{N,\tau_N} \int_{\mathrm{Herm}(2)} e^{-\frac{N}{2\tau_N^2}\operatorname{Tr} Y^2 + i\operatorname{Tr}(YS)}\, dY.$$

Now we can integrate over the Grassmann variables and obtain

$$D_{N,E}(p) = C_{N,\tau_N} \int_{\mathrm{Herm}(2)} e^{-\frac{N}{2\tau_N^2}\operatorname{Tr} Y^2} \det\big(C \otimes I_2 + iI_d \otimes Y\big)\, dY. \tag{3.5}$$

It remains to evaluate the determinant. Put

$$A_E(Y) = (B_N - E) \otimes I_2 + iI_N \otimes Y, \qquad Q_u z = u \otimes z, \quad z \in \mathbb{C}^2.$$

The range of $V_u \otimes I_2$ is the orthogonal complement of $Q_u\mathbb{C}^2$. The finite-dimensional compression determinant identity

$$\det\big((V_u \otimes I_2)^* A_E(Y)(V_u \otimes I_2)\big) = \det A_E(Y)\, \det\big(Q_u^* A_E(Y)^{-1} Q_u\big)$$

therefore applies whenever $A_E(Y)$ is invertible. Since $A_E(Y)$ is diagonal in the site index,

$$\det A_E(Y) = \prod_i \det(\tilde b_i I_2 + iY), \qquad Q_u^* A_E(Y)^{-1} Q_u = \sum_i p_i (\tilde b_i I_2 + iY)^{-1}.$$

Substituting this into (3.5) gives (3.4). □

3.3 Final representation

To simplify formulas below we set

$$c_N = N\tau_N^{-2}. \tag{3.6}$$

The positive quadratic term in (3.2) can be linearized as

$$e^{\frac{c_N}{2}(\sum_i \tilde{b}_i p_i)^2} = \left(\frac{c_N}{2\pi}\right)^{1/2} \int_{\mathbb{R}} e^{-c_N r^2/2 + c_N r \sum_i \tilde{b}_i p_i}\, dr. \tag{3.7}$$

Together with the remaining term $e^{-c_N \sum_i \tilde{b}_i^2 p_i}$ it gives

$$e^{-c_N \sum_i (\tilde{b}_i^2 - r\tilde{b}_i) p_i - c_N r^2/2}.$$

Now to replace the simplex by positive independent variables, we use the following representation of the δ-function

$$\delta\left(\sum_i p_i - 1\right) = \frac{c_N}{2\pi i} \int_\Gamma e^{c_N \zeta (1 - \sum_i p_i)} d\zeta \tag{3.8}$$

with a vertical line Γ from $\zeta_* - i\infty$ to $\zeta_* + i\infty$ which we will choose later. Then the KR measure can be rewritten as

$$d\mu^{KR}_{N,E}(p) \propto \left(\int_{\mathrm{Herm}(2)} \int_{\mathbb{R}} \int_\Gamma e^{-c_N(r^2/4 - \zeta + \mathrm{Tr}\, Y^2/2) - c_N \sum_i p_i \left((\tilde{b}_i - r/2)^2 + \zeta \right)} \right.$$
$$\left. \times \prod_i \det(\tilde{b}_i I_2 + iY) \det\left[\sum_i p_i (\tilde{b}_i I_2 + iY)^{-1} \right] dY\, dr\, d\zeta \right) dp,$$

where we have used that

$$\tilde{b}_i^2 - r\tilde{b}_i = (\tilde{b}_i - r/2)^2 - r^2/4,$$

and that

$$\sum_i r^2 p_i/4 = r^2/4$$

on a simplex Δ_{N-1}.

Changing $Y = W^* \hat{Y} W$ with 2×2 unitary W, and $\hat{Y} = \mathrm{diag}\{y_1, y_2\}$ with real y_1, y_2, and using $dY \propto (y_1 - y_2)^2 dy_1 dy_2 dW$, we get further after integrating over dW

$$d\mu^{KR}_{N,E}(p) \propto \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}} \int_\Gamma e^{c_N(\mathcal{L}_N(y_1) + \mathcal{L}_N(y_2) + \Phi_N(\bar{p}, \zeta, r))} \right.$$
$$\left. \times\, (y_1 - y_2)^2 \prod_{\alpha=1}^{2} h(\bar{p}, y_\alpha) dy_1 dy_2 dr d\zeta \right) dp, \tag{3.9}$$

with

$$\Phi_N(\bar{p}, \zeta, r) = -\sum p_j ((\tilde{b}_j - r/2)^2 + \zeta) + \zeta - r^2/4 \tag{3.10}$$
$$\mathcal{L}_N(y) = c_N^{-1} \sum_j \log(\tilde{b}_j + iy) - y^2/2,$$
$$h(\bar{p}, y) = \sum_j p_j (\tilde{b}_j + iy)^{-1}.$$

Let

$$a_i(\zeta, r) = (\tilde{b}_i - r/2)^2 + \zeta.$$

If we factor $e^{-c_N \sum_i a_i(\zeta,r)p_i} dp$ in p_i to get independent exponential measures, we get the scalar action

$$\mathcal{F}_N(r, \zeta, \bar{y}) = -\frac{r^2}{4} + \zeta - c_N^{-1} \sum_{i=1}^{N} \log a_i(r, \zeta) \tag{3.11}$$

multiplied by c_N.

More precisely, if

$$P_i^{(r,\zeta)} \sim \mathrm{Exp}(c_N a_i(r, \zeta))$$

are independent, the positive-orthant integral is factored exactly by

$$\int_{\mathbb{R}_+^N} e^{-c_N \sum_i a_i(r,\zeta)p_i} F(p)\, dp = \prod_{i=1}^{N} \frac{1}{c_N a_i(r, \zeta)} \mathbb{E}_{r,\zeta}[F(p)],$$

where $\mathbb{E}_{r,\zeta}$ is an expectation with respect all $P_i^{(r,\zeta)}$. Since $a_i(\zeta, r)$ is generally complex, this is formal notation rather than an actual probability expectation.

After absorbing c_N^{-N} into the normalization, this gives

$$\begin{aligned} &\int F(p) d\mu_{N,E}^{KR}(p) \\ &\propto \int_{\mathbb{R}^2} \int_{\mathbb{R}} \int_{\Gamma} e^{c_N \Xi_N^{(1)}(r,\zeta,y)} (y_1 - y_2)^2 \mathbb{E}_{r,\zeta} \left[F(p) \prod_{\alpha=1}^{2} h(\bar{p}, y_\alpha) \right] dy_1 dy_2 dr d\zeta \end{aligned} \tag{3.12}$$

with

$$\Xi_N^{(1)}(r, \zeta, y) = \mathcal{F}_N(r, \zeta) + \mathcal{L}_N(y_1) + \mathcal{L}_N(y_2). \tag{3.13}$$

Our aim is to prove that for the specific classes of observables we are interested in the leading order of the integral can be evaluated by computing $\mathbb{E}_{r,\zeta}$ and $h(\bar{p}, y_\alpha)$ at the saddle-points of $\mathcal{F}_N(r, \zeta)$ and $\mathcal{L}(y)$.

4 Saddle point analysis

The saddle estimates below use only the local regularity event Ω_N^{loc}. In the proofs of the limit theorems, we work on the event $\Omega_N = \Omega_N^{\mathrm{loc}} \cap \Omega_N^{(2)}$.

4.1 Saddle points

The saddle equations for $\mathcal{F}_N$ from (3.11) are

$$1 = c_N^{-1} \sum_j \frac{1}{(\tilde{b}_j - r/2)^2 + \zeta}, \qquad r/2 = c_N^{-1} \sum_j \frac{\tilde{b}_j - r/2}{(\tilde{b}_j - r/2)^2 + \zeta}. \tag{4.1}$$

By (2.3), it has a real solution

$$r_* = -2t_N, \qquad \zeta_* = \eta_N^2. \tag{4.2}$$

Take now ζ contour to be $\zeta_* + i\mathbb{R}$. Notice that since $\zeta_* > 0$, we have $\Re a_i(r, \zeta) > 0$, and so the p-integral is absolutely convergent.

The saddle equation for $\mathcal{L}_N$ from (3.10) is

$$c_N^{-1} \sum_i \frac{i}{\tilde{b}_i + iy} = y,$$

and, again by (2.3), the complex solutions are $\pm\eta_N - it_N$. One can easily check that y-contours can be moved to $-it_N + \mathbb{R}$, so the saddle points for y become

$$y_\pm = \pm\eta_N. \tag{4.3}$$

It is also useful to define

$$\widehat{\mathcal{L}}_N(y) := \mathcal{L}_N(y - it_N), \qquad \widehat{h}(p, y) := h(p, y - it_N) \tag{4.4}$$

and redefine $\Xi_N^{(1)}$ of (3.13) as

$$\Xi_N(r, \zeta, y) = \mathcal{F}_N(r, \zeta) + \widehat{\mathcal{L}}_N(y_1) + \widehat{\mathcal{L}}_N(y_2). \tag{4.5}$$

Lemma 4.1. *On the contours $\zeta \in \{\zeta_* + it : t \in \mathbb{R}\}$, $r \in \mathbb{R}$, the function $\Re\mathcal{F}(\zeta, r)$ has a unique global maximum at $r = r_*$, $t = 0$.*

Similarly, the function $\Re\widehat{\mathcal{L}}_N(y)$, $y \in \mathbb{R}$ has global maxima attained precisely at $y = y_\pm$.

Proof. Since (ζ_*, r_*) is the solution of (4.1), it is an extremum point of $\Re\mathcal{F}(\zeta, r)$ (we will check that it is maximum below in Lemma 4.2).

Now let us consider

$$\Re\mathcal{F}(\zeta_* + it, r) - \Re\mathcal{F}(\zeta_*, r_*) = -\frac{r^2}{4} + t_N^2 - \frac{1}{2c_N} \sum_i \log \frac{((\tilde{b}_i - r/2)^2 + \eta_N^2)^2 + t^2}{(\beta_i^2 + \eta_N^2)^2},$$

where $\beta_i = \tilde{b}_i + t_N(E)$.

For a fixed r, this function has a maximum at $t = 0$, so we can study

$$F_N(r) = -\frac{r^2}{4} + t_N^2 - \frac{1}{c_N} \sum_i \log \frac{(\tilde{b}_i - r/2)^2 + \eta_N^2}{\beta_i^2 + \eta_N^2} \tag{4.6}$$

It is easy to see

$$F_N''(r) = -\frac{1}{2} + \frac{1}{2c_N} \sum_i \frac{1}{(\tilde{b}_i - r/2)^2 + \eta_N^2} - \frac{1}{c_N} \sum_i \frac{\eta_N^2}{((\tilde{b}_i - r/2)^2 + \eta_N^2)^2}.$$

Notice that for any fixed $R > 0$ Proposition A.1 and (A.2), (A.6) imply

$$\sup_{|r| \le R, c\tau_N^2 < \eta \le C\tau_N^2} \Big| \frac{1}{2c_N} \sum_i \frac{1}{(\tilde{b}_i - r/2)^2 + \eta^2} - \frac{1}{c_N} \sum_i \frac{\eta^2}{\big((\tilde{b}_i - r/2)^2 + \eta^2\big)^2} \Big| \le O(\tau_N^2 + \varepsilon_N),$$

and so

$$F_N''(r) = -1/2 + o(1) < 0$$

for $|r| < R$, and hence $F_N(r)$ has a unique maximum on $|r| < R$ for any fixed R.

It remains to control $|r| > R$ for big enough R. Using the elementary inequality

$$\frac{x^2 + \eta^2}{(x - h)^2 + \eta^2} \le 2\left(1 + \frac{h^2}{\eta^2}\right), \qquad x, h \in \mathbb{R},$$

applied with $x = \beta_i$ and $h = r/2 + t_N$, we get

$$F_N(r) \le -\frac{r^2}{4} + t_N^2 + \tau_N^2 \log\Big[2\Big(1 + \frac{(r/2+t_N)^2}{\eta_N^2}\Big)\Big].$$

Because $\eta_N \asymp \tau_N^2$, one may choose a deterministic R and then N sufficiently large so that the right-hand side is at most $-cr^2$ for every $|r| \ge R$. This finishes the proof of the first statement of the lemma.

To check the statement about $\Re\widehat{\mathcal{L}}_N(y)$, $y \in \mathbb{R}$, consider

$$\Re\widehat{\mathcal{L}}_N'(y) = y\left(c_N^{-1}\sum_i \frac{1}{\beta_i^2 + y^2} - 1\right).$$

Since the function in parentheses is monotonic in y, the extremums for $y > 0$ and for $y < 0$ are unique. □

Denote $F|_*$ the value of a function at a saddle point (r_*, ζ_*, y_*) where y_* means one of the points $y_\pm$. We will need below

Lemma 4.2. *In the notations above we have*

$$-\frac{\partial^2 \mathcal{F}_N}{\partial\zeta^2}\Big|_* = -\frac{\tau_N^{-4}}{2(\pi\rho(E))^2}(1+o(1)), \quad \frac{\partial^2\mathcal{F}_N}{\partial r^2}\Big|_* = -1/2 + o(1),$$
$$\frac{\partial^2\mathcal{F}_N}{\partial\zeta\partial r}\Big|_* = o(\tau_N^{-2}), \quad \frac{\partial^2\widehat{\mathcal{L}}_N(y)}{\partial y^2}\Big|_* = -1 + o(1).$$

In addition, for $\sigma_1, \sigma_2 \in \{+1, -1\}$, make the change

$$\zeta = \zeta_* + \frac{i\tau_N^3}{\sqrt{N}}\tilde\zeta, \quad r = r_* + \frac{\tau_N}{\sqrt{N}}\tilde r, \quad y_\alpha = \sigma_\alpha\eta_N + \frac{\tau_N}{\sqrt{N}}\tilde y_\alpha, \quad \alpha = 1,2, \tag{4.7}$$

and set

$$U_N^{\sigma_1,\sigma_2} := \{(\zeta, r, y_1, y_2) : |\tilde\zeta| + |\tilde r| + |\tilde y_1| + |\tilde y_2| \le \log N\}, \qquad U_N := \bigcup_{\sigma_1,\sigma_2\in\{+1,-1\}} U_N^{\sigma_1,\sigma_2}. \tag{4.8}$$

then uniformly in U_N we have

$$\left|\frac{\partial^3\mathcal{F}_N}{\partial\tilde\zeta^3}\right| + \left|\frac{\partial^3\mathcal{F}_N}{\partial\tilde\zeta^2\partial\tilde r}\right| + \left|\frac{\partial^3\mathcal{F}_N}{\partial\tilde\zeta\partial\tilde r^2}\right| + \left|\frac{\partial^3\mathcal{F}_N}{\partial\tilde r^3}\right| + \left|\frac{\partial^3\widehat{\mathcal{L}}_N}{\partial\tilde y^3}\right| \le C\frac{\tau_N}{N^{3/2}}. \tag{4.9}$$

Moreover, in $(\tilde\zeta, \tilde r, \tilde y_1, \tilde y_2)$ variables we have

$$F_* = -c_N Hess(\mathcal{F}_N)_{\tilde\zeta,\tilde r}\Big|_* = \begin{pmatrix} c_0 + o(1) & o(1) \\ o(1) & 1/2 + o(1)\end{pmatrix} \Longrightarrow F_*^{-1} = \begin{pmatrix} c_0^{-1} + o(1) & o(1) \\ o(1) & 2 + o(1)\end{pmatrix}$$
$$L_* = -c_N(\widehat{\mathcal{L}}_N'')_{\tilde y}\Big|_* = 1 + o(1) \Longrightarrow L_*^{-1} = 1 + o(1)$$

with $c_0 = \frac{1}{2(\pi\rho(E))^2}$. Note also that in U_N

$$(y_1 - y_2)^2 = \begin{cases} 4\eta_N^2(1 + O((\tau_N^2 N)^{-1/2}\log N)), & y_1 = \pm\eta_N + \frac{\tau_N}{\sqrt{N}}\tilde y_1, \quad y_2 = \mp\eta_N + \frac{\tau_N}{\sqrt{N}}\tilde y_2, \\ \eta_N^2 \cdot O((\tau_N^2 N)^{-1}\log^2 N) & y_1 = \pm\eta_N + \frac{\tau_N}{\sqrt{N}}\tilde y_1, \quad y_2 = \pm\eta_N + \frac{\tau_N}{\sqrt{N}}\tilde y_2. \end{cases} \tag{4.10}$$

and therefore for Ξ_N of (4.5) we get an expansion

$$c_N(\Xi_N - \Xi_{N,*}) = -\frac{1}{2}(F_*\Delta_1, \Delta_1) - \frac{1}{2}L_*(\Delta_2, \Delta_2) + O(\log^3 N(N\tau_N^2)^{-1/2}), \tag{4.11}$$

where $\Delta_1 = (\tilde\zeta, \tilde r)^t$, $\Delta_2 = (\tilde y_1, \tilde y_2)^t$.

Proof. Denote $\beta_i = \tilde b_i + t_N$. Computing the derivatives and using Proposition A.1 and Lemma A.1 we get

$$\begin{aligned}
&\frac{\partial^2 \mathcal{F}_N}{\partial\zeta^2}\Big|_* = \frac{\tau_N^2}{N}\sum_i \frac{1}{a_i(r,\zeta)^2}\Big|_* = \frac{\tau_N^2}{N}\sum_i \frac{1}{(\beta_i^2+\eta_N^2)^2} = \frac{\frac12 + o(1)}{\eta_N^2} \propto \tau_N^{-4},\\
&\frac{\partial^2 \mathcal{F}_N}{\partial r^2}\Big|_* = \frac{\tau_N^2}{N}\sum_i \frac{\beta_i^2}{(\beta_i^2+\eta_N^2)^2} - 1 = -1/2 + o(1),\\
&\frac{\partial^2 \mathcal{F}_N}{\partial\zeta\partial r}\Big|_* = -\frac{\tau_N^2}{N}\sum_i \frac{\beta_i}{(\beta_i^2+\eta_N^2)^2} = o(\tau_N^{-2}),\\
&\frac{\partial^2 \widehat{\mathcal{L}}_N(y)}{\partial y^2}\Big|_* = -\frac{\tau_N^2}{N}\sum_i \frac{2\eta_N^2}{(\beta_i^2+\eta_N^2)^2} \pm \frac{i\tau_N^2}{N}\sum_i \frac{2\eta_N\beta_i}{(\beta_i^2+\eta_N^2)^2} = -1 + o(1),
\end{aligned}$$

which proves the first statement and the expression for the Hessian after the change of variables. The third derivatives are

$$\begin{aligned}
&\frac{\partial^3 \mathcal{F}_N}{\partial\zeta^3} = -\frac{\tau_N^2}{N}\sum_i \frac{2}{a_i(r,\zeta)^3}\\
&\frac{\partial^3 \mathcal{F}_N}{\partial r^3} = -\frac{3\tau_N^2}{2N}\sum_i \frac{(\tilde b_i - r/2)}{a_i(r,\zeta)^2} + \frac{2\tau_N^2}{N}\sum_i \frac{(\tilde b_i - r/2)^3}{a_i(r,\zeta)^3},\\
&\frac{\partial^3 \mathcal{F}_N}{\partial\zeta^2\partial r} = \frac{\tau_N^2}{N}\sum_i \frac{2(\tilde b_i - r/2)}{a_i(r,\zeta)^3},\\
&\frac{\partial^3 \mathcal{F}_N}{\partial\zeta\partial r^2} = \frac{\tau_N^2}{2N}\sum_i \frac{1}{a_i(r,\zeta)^2} - \frac{2\tau_N^2}{N}\sum_i \frac{(\tilde b_i - r/2)^2}{a_i(r,\zeta)^3},\\
&\frac{\partial^3 \Re\widehat{\mathcal{L}}_N(y)}{\partial y^3} = -\frac{\tau_N^2}{N}\sum_i \frac{6y}{((\tilde b_i + t_N)^2 + y^2)^2} + \frac{\tau_N^2}{N}\sum_i \frac{8y^3}{((\tilde b_i + t_N)^2 + y^2)^3},\\
&\frac{\partial^3 \Im\widehat{\mathcal{L}}_N(y)}{\partial y^3} = -\frac{2\tau_N^2}{N}\sum_i \frac{(\tilde b_i + t_N)}{((\tilde b_i + t_N)^2 + y^2)^2} + \frac{\tau_N^2}{N}\sum_i \frac{8(\tilde b_i + t_N)y^2}{((\tilde b_i + t_N)^2 + y^2)^3}
\end{aligned}$$

Using

$$\Big|\frac{1}{a_i(r,\zeta)}\Big| \le \frac{1}{\eta_N^2}, \quad \Big|\frac{(\tilde b_i - r/2)}{a_i(r,\zeta)}\Big| \le \frac{|\tilde b_i - r/2|}{(\tilde b_i - r/2)^2 + \eta_N^2} \le \frac{2}{\eta_N},$$

similar inequalities with y, and Proposition A.1, we obtain

$$\begin{aligned}
&\frac{\tau_N^2}{N}\sum_i \Big|\frac{1}{a_i(r,\zeta)^3}\Big| \le \eta_N^{-4}\cdot\frac{\tau_N^2}{N}\sum_i \frac{1}{(\tilde b_i - r/2)^2 + \eta_N^2} = O(\eta_N^{-4}) = O(\tau_N^{-8}),\\
&\frac{\tau_N^2}{N}\sum_i \Big|\frac{(\tilde b_i - r/2)}{a_i(r,\zeta)^2}\Big| \le \frac{2}{\eta_N}\cdot\frac{\tau_N^2}{N}\sum_i \frac{1}{(\tilde b_i - r/2)^2 + \eta_N^2} = O(\eta_N^{-1}) = O(\tau_N^{-2}).
\end{aligned}$$

Other terms can be bounded similarly. Changing to $(\tilde\zeta, \tilde r, \tilde y_1, \tilde y_2)$ variables, we get (4.9). □

4.2 Proofs of Theorems 2.1-2.4

Proof of Theorem 2.1. It is easy to see that to prove (2.11), it is sufficient to prove that

$$\int I_q^s \, d\mu_{N,E}^{KR} = \Big(\Gamma(q+1)\sum_{i=1}^{N} m_i(E)^q\Big)^s (1+o(1)), \quad s=1,2 \tag{4.12}$$

where $\mu_{N,E}^{KR}$ was defined in (3.9). Denote

$$\langle \dots \rangle = \int (\dots)(y_1-y_2)^2 \exp\{c_N \Xi_N(\zeta,r,y)\} d\zeta dr dy_1 dy_2 \tag{4.13}$$

where $\Xi(\zeta,r,y)$ is defined in (4.5).

Let $\mathcal{Z}$ be a normalizing factor in (3.9), i.e.

$$\mathcal{Z} = \int dp d\zeta dr dy_1 dy_2 \prod_{l=1}^{2} \widehat{h}(\bar{p}, y_l) \tag{4.14}$$

$$\times\, e^{c_N(\widehat{\mathcal{L}}_N(y_1)+\widehat{\mathcal{L}}_N(y_2)+\Phi_N(\bar{p},\zeta,r))}(y_1-y_2)^2. \tag{4.15}$$

Integrating with respect to $d\bar{p}$, we get

$$\mathcal{Z} = \langle g(\zeta,r,y_1) g(\zeta,r,y_2) + \Sigma' \rangle$$

where

$$g(\zeta,r,y) = c_N^{-1} \sum_j \frac{(\tilde{b}_j + t_N + iy)^{-1}}{(\tilde{b}_j - r/2)^2 + \zeta} \tag{4.16}$$

$$\Sigma' = c_N^{-2} \sum_j \frac{(\tilde{b}_j + t_N + iy_1)^{-1}(\tilde{b}_j + t_N + iy_2)^{-1}}{((\tilde{b}_j - r/2)^2 + \zeta)^2}$$

Notice that due to Lemmas 4.1 – 4.2 by the standard saddle-point analysis argument the contribution of the U_N^c to the integral is bounded by $e^{-C\log^2 N}$. Indeed, in each of the four saddle neighbourhoods in (4.8), Taylor expansion and (4.9) give

$$c_N \Re(\Xi_N - \Xi_{N,*}) \le -c(\tilde{\zeta}^2 + \tilde{r}^2 + \tilde{y}_1^2 + \tilde{y}_2^2) + O\Big((N\tau_N^2)^{-1/2}(|\tilde{\zeta}| + |\tilde{r}| + |\tilde{y}_1| + |\tilde{y}_2|)^3\Big).$$

Hence, on the boundary of U_N, the real part of the exponent is smaller than its saddle value by $c\log^2 N$. Lemma 4.1 propagates this gap outside a fixed neighbourhood: the real part decreases in $|\Im\zeta|$, strict concavity controls the r-direction, and the monotonicity assertion for $\Re\widehat{\mathcal{L}}_N(y)$ controls the two y-directions. The explicit tail bounds in the proof of that lemma treat the unbounded portions of the contours. Consequently, the contribution of U_N^c to the saddle integral with any polynomially bounded prefactor is $O(e^{-c\log^2 N})$.

In addition, by using Lemmas A.1-A.2 and Proposition A.1 it is easy to get w.h.p that inside U_N of (4.8)

$$|g(\zeta_*, r_*, y_*)| \asymp \eta_N^{-1}, \quad |\Sigma'| \le C/N\eta_N^3, \tag{4.17}$$

so

$$|\Sigma'| \le C|g(\zeta_*,r_*,y_*)|^2(\tau_N^2 N)^{-1} \Rightarrow \mathcal{Z} = \langle g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle(1+o(1))$$

Similarly, integrating by $\bar{p}$ in (4.12) we get

$$\int I_q^s \, d\mu_{N,E}^{KR} = \frac{\langle f_q^s(\zeta,r) g(\zeta,r,y_1) g(\zeta,r,y_2)\rangle}{\langle g(\zeta,r,y_1) g(\zeta,r,y_2)\rangle}(1+O((\tau_N^2 N)^{-1})),\ s=1,2,$$

where

$$f_q(\zeta,r) = \Gamma(q+1) c_N^{-q} \sum_j \frac{1}{((\tilde{b}_j - r/2)^2+\zeta)^q}. \tag{4.18}$$

Let us prove the following simple lemma:

Lemma 4.3. *Let the function $f(\zeta,r,y)$ uniformly in U_N satisfies the relations:*

$$f = f_*(1+o(1)), \quad f_* = f(\zeta,r,y)\Big|_* \tag{4.19}$$

Then

$$\langle f\rangle = \langle f_*\rangle (1+o(1)). \tag{4.20}$$

The sufficient conditions for (4.19) are

$$\Big|\frac{\partial f}{\partial \zeta}\Big| \le |f_*| \tau_N^{-4} \nu_N, \quad \Big|\frac{\partial f}{\partial r}\Big| \le |f_*| \tau_N^{-2} \nu_N, \quad \Big|\frac{\partial f}{\partial y_{1,2}}\Big| \le |f_*| \tau_N^{-2} \nu_N \tag{4.21}$$

uniformly in U_N, where $\{\nu_N\}$ is some sequence growing slowly than any degree of N.

Proof. The statement of the lemma is trivial for positive measures. However, since the measure $< ... >$ is not positive, we need to show the relation

$$\frac{\int \exp\{c_N \Re \Xi_N\} |y_1-y_2|^2 d\zeta dr dy|}{\Big|\int \exp\{c_N \Xi_N\}(y_1-y_2)^2 d\zeta dr dy\Big|} = \frac{|(\Re L_*)^2 \det \Re F_*|^{-1/2}}{|L_*^2 \det F_*|^{-1/2}}(1+o(1)) \le C,$$

if we choose the same $y_\pm$ for y_1 and y_2 in U_N, and

$$\frac{\int \exp\{c_N \Re \Xi_N\} d\zeta dr dy|}{\Big|\int \exp\{c_N \Xi_N\} d\zeta dr dy\Big|} = \frac{|(\Re L_*)^4 \det \Re F_*|^{-1/2}}{|L_*^4 \det F_*|^{-1/2}}(1+o(1)) \le C,$$

if we choose different $y_\pm$ for y_1 and y_2 (see (4.10)).

Note that both inequalities easily follow from (4.5), (4.11), and saddle point computations of Lemma 4.2.

To check that (4.21) implies (4.19) we use the change (4.8) for f. Indeed, denoting ∇f the vector of derivatives of f with respect to the variable $(\tilde{\zeta}, \tilde{r}, \tilde{y}_1, \tilde{y}_2)$ we get from (4.21)

$$|\nabla f| \le C |f_*| \nu_N / (\tau_N^2 N)^{1/2},$$

which implies (4.19) after the trivial Taylor expansion.

□

Using the lemma, we conclude that to finish the proof of Theorem 2.1, it suffices to check conditions (4.21) for $f = g(\zeta,r,y_1)g(\zeta,r,y_2)$ and for $f = f_q^s(\zeta,r) g(\zeta,r,y_1) g(\zeta,r,y_2)$, $s=1,2$.

We first verify the saddle-neighbourhood estimates for f_q directly. Set

$$a_j(r,\zeta) = (\widetilde{b}_j - r/2)^2 + \zeta, \qquad a_{j,*} = \beta_j^2 + \eta_N^2, \qquad \beta_j = \widetilde{b}_j + t_N.$$

Writing $\Delta r = r - r_*$ and $\Delta\zeta = \zeta - \eta_N^2$, we have

$$a_j(r,\zeta) - a_{j,*} = -\beta_j \Delta r + \frac{(\Delta r)^2}{4} + \Delta\zeta.$$

Since

$$\frac{|\beta_j|}{\beta_j^2 + \eta_N^2} \le \frac{1}{2\eta_N}, \qquad \frac{1}{\beta_j^2 + \eta_N^2} \le \eta_N^{-2},$$

the definition of U_N implies, uniformly in j and in U_N,

$$\left| \frac{a_j(r,\zeta) - a_{j,*}}{a_{j,*}} \right| \le C \left(\frac{|\Delta r|}{\eta_N} + \frac{|\Delta r|^2 + |\Delta\zeta|}{\eta_N^2} \right) \le C \frac{\log N}{\sqrt{N\tau_N^2}} =: d_N = o(1).$$

Thus, for every fixed $q > 0$,

$$a_j(r,\zeta)^{-q} = a_{j,*}^{-q}(1 + O_q(d_N))$$

uniformly in j and in U_N. Hence,

$$f_q(r,\zeta) = f_{q,*}(1 + O_q(d_N)), \qquad f_{q,*} = \Gamma(q+1) c_N^{-q} \sum_j a_{j,*}^{-q} > 0.$$

The same comparison yields

$$|\partial_\zeta f_q| \le C_q \eta_N^{-2} f_{q,*}, \qquad |\partial_r f_q| \le C_q \eta_N^{-1} f_{q,*}.$$

Together with the corresponding estimates for g, these bounds verify the conditions of Lemma 4.3 for

$$g(\zeta, r, y_1) g(\zeta, r, y_2), \qquad f_q(r,\zeta)^s g(\zeta, r, y_1) g(\zeta, r, y_2), \quad s = 1, 2.$$

Note also that because of (4.10), the leading contribution is given by the saddle-points $y_{1,*} = \pm\eta_N, \quad y_{2,*} = \mp\eta_N$. Indeed the $\langle .. \rangle$ measure contains $(y_1 - y_2)^2$. It is of order η_N^2 for $y_{1,*} = \pm\eta_N, \quad y_{2,*} = \mp\eta_N$ and of order τ_N^2/N for $y_{1,*} = \pm\eta_N, \quad y_{2,*} = \pm\eta_N$ on the Gaussian fluctuation scale, and the other functions with respect to y (i.e., h, $\mathcal{L}$) just have a complex conjugated values at $y = \pm\eta_N$. Therefore, the total same-sign contribution is $O(K_N^{-1})$ relative to the mixed-sign one.

It remains to check (2.12) – (2.14). If $q > 1/2$, then (A.11) for $F_q(u) = (1+u^2)^{-q}$ yields

$$K_N^{q-1} \sum_i m_i^q = \frac{1}{K_N} \sum_i F_q(\tfrac{\tilde{b}_i + t_N}{\eta_N}) = \frac{N}{K_N} \Big(\int F_q(\tfrac{b - E + t_N}{\eta_N}) \rho(b) db + o(\eta_N) \Big).$$

Changing $u = \frac{b - E + t_N}{\eta_N}$, we get

$$\eta_N(E) \int F(u) \rho(E - t_N(E) + \eta_N(E) u)\, du = \eta_N(E) \rho(E) \int F(u)\, du\, [1 + o(1)]$$

uniformly in E. Since

$$\frac{N \eta_N \rho(E)}{K_N(E)} = \frac{\tau_N^2 \rho(E)}{\eta_N} = \frac{1}{\pi} + o(1),$$

we get

$$K_N^{q-1}\sum_i m_i^q \longrightarrow \frac{1}{\pi}\int_{\mathbb{R}}\frac{du}{(1+u^2)^q} = \frac{\Gamma(q-1/2)}{\sqrt{\pi}\,\Gamma(q)}.$$

This proves (2.12).

For $0 < q < 1/2$, fix $h > 0$ and split at $|b_i - E| = h$. On the complement the function $|x-E|^{-2q}$ is bounded and we get similarly

$$N^{\gamma q-1}\sum_{|b_i-E|>h} m_i^q \longrightarrow \int_{|x-E|>h}|x-E|^{-2q}\rho(x)\,dx.$$

The remaining part is bounded, uniformly in N, by $C\int_{|x-E|\le 2h}|x-E|^{-2q}\rho(x)dx + o(1)$, which tends to zero as $h \downarrow 0$. This proves (2.14).

For $q = 1/2$, the $s = 1/2$ case of (A.8), applied at $r = -2t_N$ and $\zeta = \eta_N^2$, gives

$$\frac{1}{N}\sum_{j=1}^N \frac{1}{\sqrt{(b_j - E + t_N)^2 + \eta_N^2}} = \int_{\mathbb{R}}\frac{\rho(b)\,db}{\sqrt{(b-E+t_N)^2+\eta_N^2}} + O(\varepsilon_N).$$

By the smoothness of ρ near I and the fact that $t_N = o(1)$, uniformly in $E \in I$,

$$\int_{\mathbb{R}}\frac{\rho(b)\,db}{\sqrt{(b-E+t_N)^2+\eta_N^2}} = 2\rho(E)\log\frac{1}{\eta_N} + o\Big(\log\frac{1}{\eta_N}\Big).$$

Since

$$K_N^{-1/2}\sum_{j=1}^N m_j^{1/2} = \frac{\tau_N^2}{N\eta_N}\sum_{j=1}^N\frac{1}{\sqrt{(b_j-E+t_N)^2+\eta_N^2}}, \qquad \frac{\tau_N^2\rho(E)}{\eta_N} = \frac{1}{\pi}[1+o(1)],$$

we obtain

$$K_N^{-1/2}\sum_{j=1}^N m_j^{1/2} = \frac{2}{\pi}\log\frac{1}{\eta_N}\,[1+o(1)].$$

This proves (2.13).

□

Proof of Theorem 2.2. It is easy to see that to prove the theorem it is sufficient to prove relations

$$\begin{aligned}
&\mathbb{E}\big[N^{-f(\alpha)}\mathcal{N}_N(\alpha)\big] = \sqrt{\pi}\rho(E) + o(1), \quad 2-\gamma < \alpha < \gamma, \\
&\mathbb{E}\big[N^{-f(\alpha)}\mathcal{N}_N(\alpha)\big] = \int \rho(b)e^{-(b-E)^2}db + o(1), \quad \alpha = \gamma, \\
&\mathbb{E}\big[N^{-f(\alpha)}(N - \mathcal{N}_N(\alpha))\big] = \mu_2(E) + o(1), \quad \gamma + 1 > \alpha > \gamma, \\
&\mathrm{Var}\big[\mathcal{N}_N(\alpha)N^{-f(\alpha)}\big] = o(1),
\end{aligned} \tag{4.22}$$

for

$$\mathcal{N}_N(\alpha) = \#\{i : p_i \ge N^{-\alpha}\}.$$

Here $\mu_2(E)$ is defined in (2.2).

Indeed, let $2-\gamma<\alpha<\gamma$ and ε is small enough. Write

$$\frac{\log \mathcal{N}_N(\alpha,\varepsilon)}{\log N}=\frac{\log\Big(\mathcal{N}_N(\alpha+\varepsilon)-\mathcal{N}_N(\alpha-\varepsilon)\Big)}{\log N}=\frac{\log \mathcal{N}_N(\alpha+\varepsilon)+\log\Big(1-\frac{\mathcal{N}_N(\alpha-\varepsilon)}{\mathcal{N}_N(\alpha+\varepsilon)}\Big)}{\log N}. \quad (4.23)$$

Fix small $\delta>0$. The relation (4.22) implies

$$\mathcal{N}_N(\alpha\pm\varepsilon)/N^{f(\alpha\pm\varepsilon)}\xrightarrow{p}\sqrt{\pi}\rho(E)$$
$$\Longrightarrow \mathbb{P}\Big(\tfrac{\log(\sqrt{\pi}\rho(E)-\delta)}{\log N}\le \tfrac{\log \mathcal{N}_N(\alpha+\varepsilon)}{\log N}-f(\alpha+\varepsilon)\le \tfrac{\log(\sqrt{\pi}\rho(E)+\delta)}{\log N}\Big)\to 1,$$

and so the first term of (4.23) goes to $f(\alpha+\varepsilon)$, as $N\to\infty$. Similarly, since $f(\alpha+\varepsilon)-f(\alpha-\varepsilon)=\varepsilon$,

$$\mathbb{P}\Big(\tfrac{1-\delta}{N^{f(\alpha+\varepsilon)-f(\alpha-\varepsilon)}}\le \tfrac{\mathcal{N}_N(\alpha-\varepsilon)}{\mathcal{N}_N(\alpha+\varepsilon)}\le \tfrac{1+\delta}{N^{f(\alpha+\varepsilon)-f(\alpha-\varepsilon)}}\Big)\to 1$$

implies that the second term of (4.23) goes to 0, as $N\to\infty$. The case $\alpha>\gamma$ can be considered in a similar way using $C_N(\alpha)=N-\mathcal{N}_N(\alpha)$ instead of $\mathcal{N}_N(\alpha)$. For $\alpha=\gamma$, the estimates on the two branches give, for sufficiently small $\varepsilon>0$,

$$\mathcal{N}_N(\gamma-\varepsilon)=o_{\mathbb{P}}(N),\qquad C_N(\gamma+\varepsilon)=o_{\mathbb{P}}(N).$$

Hence,

$$\mathcal{N}_N(\gamma,\varepsilon)=N-C_N(\gamma+\varepsilon)-\mathcal{N}_N(\gamma-\varepsilon)=N[1+o_{\mathbb{P}}(1)],$$

which yields $f(\gamma)=1$.

To prove the first relation of (4.22), observe that for different j_1,j_2,j_3

$$N^{-f(\alpha)}\int d\bar p e^{N\tau_N^{-2}\Phi_N(\bar p,\zeta,r)}\frac{p_{j_1}}{\tilde b_{j_1}+iy_1}\frac{p_{j_2}}{\tilde b_{j_2}+iy_2}\theta(p_{j_3}-N^{-\alpha}) \quad (4.24)$$
$$=c_N^{-2}N^{-f(\alpha)}\frac{((\tilde b_{j_1}-r/2)^2+\zeta)^{-1}}{\tilde b_{j_1}+iy_1}\frac{((\tilde b_{j_2}-r/2)^2+\zeta)^{-1}}{\tilde b_{j_2}+iy_2}e^{-N^{\gamma-\alpha}((\tilde b_{j_3}-r/2)^2+\zeta)}\int d\bar p e^{N\tau_N^{-2}\Phi_N(\bar p,\zeta,r)}$$

Introduce the function

$$V_\alpha(\zeta,r)=N^{-f(\alpha)}\sum_j e^{-N^{\gamma-\alpha}((\tilde b_j-r/2)^2+\zeta)}. \quad (4.25)$$

For $2-\gamma<\alpha<\gamma$, by (A.9) in Ω_N we have uniformly in U_N

$$V_\alpha(\zeta,r)=N^{1-f(\alpha)}\int\rho(b)db e^{-N^{\gamma-\alpha}((b-E-r/2)^2+\zeta)}+O(N^{-1/2+(\gamma-\alpha)/4}\log N)$$
$$=\sqrt{\pi}\rho(E+r/2)N^{1-(\gamma-\alpha)/2-f(\alpha)}(1+O(N^{2-\gamma-\alpha})+O(N^{-(\gamma-\alpha)})+O(N^{-1/2+(\gamma-\alpha)/4}\log N))$$
$$=\sqrt{\pi}\rho(E)(1+O(r)+O(N^{2-\gamma-\alpha})+O(N^{-(\gamma-\alpha)}))+O(N^{-1/2+(\gamma-\alpha)/4}\log N)$$
$$=V_{\alpha,*}(1+o(1)).$$

Then the sum of terms of (4.24) with $j_3=j_1$ or $j_3=j_2$ can be estimated as $N^{-f(\alpha)}g_*^2\asymp N^{-f(\alpha)}g_*^2V_{\alpha,*}$ and the sum of terms with $j_3\ne j_1$ and $j_3\ne j_2$ was estimated in (4.17). Thus the sum of integrals with some coinciding indices is bounded by $C(N^{-f(\alpha)}+(N\tau_N^2)^{-1}g_*^2V_{\alpha,*})$.

Hence we get for $2-\gamma<\alpha<\gamma$:

$$\begin{aligned}\int \mu^{KR}_{N,E}(\bar p)d\bar p(N^{-f(\alpha)}\mathcal{N}_N(\alpha)) =&\frac{\langle V_\alpha(\zeta,r)g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle}{\langle g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle}(1+O((N\tau_N^2)^{-1})+O(N^{-f(\alpha)}))\\ &= V_{\alpha,*}(1+o(1))\end{aligned}$$

which implies the first relation of (4.22) for $2-\gamma<\alpha<\gamma$.

For $\alpha=\gamma$, $(f(\alpha)=1)$, according to (A.12) we have

$$V_\alpha(\zeta,r)=\int \rho(b)dbe^{-(b-E)^2}+O(N^{-1/2}\log N)+O(|r|)+O(|\zeta|).$$

All other estimates are the same.

For $\alpha>\gamma$ by the same argument we obtain an analogue of (4.25) for $N^{-f(\alpha)}(N-\mathcal{N}_N(\alpha))$ with

$$\begin{aligned}\widetilde V_\alpha(\zeta,r) &= N^{-f(\alpha)}\sum_j\left(1-e^{-N^{\gamma-\alpha}((\tilde b_j-r/2)^2+\zeta)}\right)\\ &= \frac{1}{N}\sum_j a_j(r,\zeta)\,\psi\big(N^{\gamma-\alpha}a_j(r,\zeta)\big)\,,\\ &= \frac{1}{N}\sum_j a_j(r,\zeta)+o(1)=\mu_2(E)+o(1),\quad \psi(z)=\frac{1-e^{-z}}{z}\end{aligned}$$

uniformly on U_N. Here the second equality follows by a standard truncation argument, using $\psi(z)\to 1$ as $z\to 0$ on bounded sets and (2.8) to control the tails.

To estimate the sum of integrals with $\theta(N^{-\alpha}-p_{j_3})$ instead of $\theta(p_{j_3}-N^{-\alpha})$ in (4.24) with $j_3=j_1$ or $j_3=j_2=j_1$ we use the relation

$$\begin{aligned}&\Big|\int_0^{N^{-\alpha}} p_j^s e^{-N^\gamma((\tilde b_j-r/2)^2+\zeta)p_j}\,dp_j\Big|\\ &\qquad\le N^{-s\alpha}\int_0^{N^{-\alpha}} e^{-N^\gamma p_j\Re((\tilde b_j-r/2)^2+\zeta)}\,dp_j,\qquad s=1,2.\end{aligned}$$

Denote by Σ_1' the sum over $j_3=j_1$ and $j_2\neq j_1$, and by Σ_2' the sum with $j_3=j_2=j_1$. Using the previous inequalities and $g_*\asymp\eta_N^{-1}$, we obtain

$$|\Sigma_1'|+|\Sigma_2'|\le CN^{1-\alpha-\gamma}|\tilde V_* g_*^2|=o(|\tilde V_* g_*^2|).$$

The sum of the terms with $j_1=j_2$ and $j_3\neq j_1$ was estimated in (4.17). Thus we get

$$\begin{aligned}&\int \mu^{KR}_{N,E}(\bar p)\,d\bar p\,N^{-f(\alpha)}(N-\mathcal{N}_N(\alpha))\\ &=\frac{\langle\tilde V_\alpha(\zeta,r)g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle}{\langle g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle}(1+O((N\tau_N^2)^{-1})+O(N^{1-\alpha-\gamma}))\\ &=\tilde V_{\alpha,*}(1+o(1)).\end{aligned}$$

Thus we obtain the third relation in (4.22).

To obtain the last relation of (4.22) we apply the same argument to V_α^2 and $\tilde V_\alpha^2$. The terms with two distinct marked indices give

$$V^2_{\alpha,*}(1+o(1)),\qquad \text{respectively}\qquad \widetilde V^2_{\alpha,*}(1+o(1)),$$

while the terms with coinciding indices are $o(1)$ by the estimates above. Thus the corresponding second moment equals the square of the first moment up to $o(1)$, proving the last relation in (4.22).

□

Proof of Theorem 2.3. Let $\varphi \in C_c^+(\mathbb{R}\times(-A,\infty))$. To prove the theorem, it is enough to show convergence of the Laplace functional

$$I_\varphi = \mathbb{E}_{\mathrm{KR}} \exp\left\{-\sum_j \varphi(v_j, K_N p_j - \alpha_N)\right\} \longrightarrow \exp\left\{-\int_{\mathbb{R}} \frac{e^{-v^2}}{\sqrt{\pi}}\, dv \int_{\mathbb{R}} e^{-x}\big(1-e^{-\varphi(v,x)}\big)\, dx\right\}. \tag{4.26}$$

Set

$$\mathcal{Q}_\varphi(r,\zeta) := \prod_{j=1}^N \left(c_N a_j(r,\zeta)\int_0^\infty e^{-c_N a_j(r,\zeta)p - \varphi(v_j, K_N p-\alpha_N)}\, dp\right),$$

where $a_j(r,\zeta) = (\tilde b_j - r/2)^2 + \zeta$. With

$$q_j(r,\zeta) := \frac{a_j(r,\zeta)}{\eta_N^2} e^{-\alpha_N a_j(r,\zeta)/\eta_N^2}\int_{-\alpha_N}^\infty e^{-a_j(r,\zeta)x/\eta_N^2}\big(1-e^{-\varphi(v_j,x)}\big)\, dx,$$

the change of variables $x = K_N p - \alpha_N$ and $c_N/K_N = \eta_N^{-2}$ give

$$\mathcal{Q}_\varphi(r,\zeta) = \prod_{j=1}^N (1-q_j(r,\zeta)). \tag{4.27}$$

Doing the same transformations with integrals

$$c_N a_j(r,\zeta)\int_0^\infty p e^{-c_N a_j(r,\zeta)p - \varphi(v_j, K_N p-\alpha_N)}\, dp,$$

we get that integrating with respect to p_i and arguing as in the proof of Theorem 2.1, we can bound the diagonal contribution by $o(|g_*|^2)$, and write

$$I_\varphi = \frac{\langle \mathcal{Q}_\varphi(r,\zeta)\hat g(\zeta,r,y_1)\hat g(\zeta,r,y_2)\rangle}{\langle g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle} + o(1)$$

$$\hat g(\zeta,r,y) = c_N^{-1}\sum_j \frac{a_j^{-1}}{(\tilde b_j + t_N + iy)}\,\frac{1 - a_j^2\eta_N^{-4}\int_{-\alpha_N}^\infty (x+\alpha_N)e^{-a_j\eta_N^{-2}(x+\alpha_N)}(1-e^{-\varphi(v_j,x)})dx}{1 - a_j\eta_N^{-2}\int_{-\alpha_N}^\infty e^{-a_j\eta_N^{-2}(x+\alpha_N)}(1-e^{-\varphi(v_j,x)})dx}$$

Choose R, T so that $\operatorname{supp}\varphi \subset [-R,R]\times[-T,T]$.

Using (A.9) and (2.15), we get

$$|g-\hat g| \le CN^{-1}\eta_N^{-2}\sum_j \int_{-T}^T dx e^{-\alpha_N}(\alpha_N + x)e^{-\eta_N^{-2}(\alpha_N+x)(\tilde b_j - r/2)^2}$$

$$\le C\eta_N^{-2}\alpha_N \int_{-T}^T dx N^{-1} e^{-\alpha_N}\sum_j e^{-\eta_N^{-2}(\alpha_N+x)(\tilde b_j - r/2)^2}$$

$$\le C\rho(E)\frac{\alpha_N}{\eta_N\sqrt{\alpha_N}}e^{-\alpha_N} \le C\alpha_N(\eta_N K_N)^{-1} \le C\alpha_N(N\eta_N^2)^{-1} \le C|g_*|\alpha_N(N\eta_N)^{-1} = |g_*|o(1)$$

$$\Rightarrow \hat g(\zeta,r,y) = g(\zeta,r,y)(1+o(1)) = g_*(1+o(1)). \tag{4.28}$$

Observe, that uniformly in U_N, $q_j = 0$ unless $|v_j| \le R$, and on these indices

$$\frac{a_j(r,\zeta)}{\eta_N^2} = 1 + \frac{v_j^2}{\alpha_N} + o(\alpha_N^{-1}).$$

Therefore,

$$\sum_j q_j(r,\zeta) = e^{-\alpha_N} \sum_{|v_j|\le R} e^{-v_j^2} \int_{\mathbb{R}} e^{-x}\big(1 - e^{-\varphi(v_j,x)}\big)\, dx\, [1 + o(1)]. \tag{4.29}$$

Notice that (A.11) with expansion of $\rho(b)$ at $b = E - t_N + v\eta_N/\sqrt{\alpha_N}$ give for general F satisfying conditions of (A.11)

$$\frac{\sqrt{\alpha_N}}{N\eta_N} \sum_j F(v_j) = \rho(E) \int_{\mathbb{R}} F(v)\, dv + o(1).$$

Although the function

$$F(v) := e^{-v^2} \int_{\mathbb{R}} e^{-x}\big(1 - e^{-\varphi(v,x)}\big)\, dx$$

need not be C^1, it is continuous and compactly supported. Hence, we can uniformly approximate it by functions in $C_c^1(\mathbb{R})$. The error in the corresponding empirical average is negligible, uniformly on $\Omega_N^{(2)}$, which can be obtained by the standard argument with application of (A.11) to the fixed non-negative cutoff $\chi \in C_c^1(\mathbb{R})$ with $\chi \ge 1$ on a fixed compact set containing the supports of F and all approximants. Therefore, (A.11) applies to such F as well.

Using this in (4.29), together with (2.15) and $\eta_N = \pi\rho(E)\tau_N^2[1 + o(1)]$, gives

$$\sum_j q_j(r,\zeta) \longrightarrow \int_{\mathbb{R}} \frac{e^{-v^2}}{\sqrt{\pi}}\, dv \int_{\mathbb{R}} e^{-x}\big(1 - e^{-\varphi(v,x)}\big)\, dx. \tag{4.30}$$

Moreover $\max_j |q_j| = O(e^{-\alpha_N}) = o(1)$, hence $\sum_j |q_j|^2 = o(1)$. Thus taking log of (4.27) and expanding it up to the second term gives, uniformly in U_N,

$$\mathcal{Q}_\varphi(r,\zeta) = \exp\left\{ -\int_{\mathbb{R}} \frac{e^{-v^2}}{\sqrt{\pi}}\, dv \int_{\mathbb{R}} e^{-x}\big(1 - e^{-\varphi(v,x)}\big)\, dx \right\} (1 + o(1)). \tag{4.31}$$

Now (4.26) can be obtained using (4.31), (4.28), and Lemma 4.3. □

Proof of Corollary 2.1. Define

$$\mathcal{M}_N(t) = \sum_j \theta(K_N p_j - \alpha_N - t) = \sum_j \theta_j \tag{4.32}$$

Fix any x and study

$$F_t(x) = \mathbb{E}_{\mathrm{KR}}[e^{-x\mathcal{M}_N(t)}].$$

To show $K_N p_{\max} - \alpha_N \Longrightarrow \mathcal{G}$, it is enough to prove that for any fixed t, x

$$F_t(x) \to e^{-(1-e^{-x})e^{-t}}. \tag{4.33}$$

Integrating with respect to $\bar{p}$ and using the same argument as in Theorem 2.1 to treat the diagonal terms, we obtain that

$$F_t(x) = \frac{\langle g(\zeta,r,y_1)g(\zeta,r,y_2)\prod(1-(1-e^{-x})e^{-\eta_N^{-2}(\alpha_N+t)((\tilde{b}_j-r/2)^2+\zeta)})\rangle}{\langle g(\zeta,r,y_1)g(\zeta,r,y_2)\rangle}(1+o(1)).$$

Since

$$\begin{aligned}\Sigma_1 =& \sum_j \log\left(1-(1-e^{-x})e^{-\eta_N^{-2}(\alpha_N+t)((\tilde{b}_j-r/2)^2+\zeta)}\right)\\ =& -(1-e^{-x})\sum_j e^{-\eta_N^{-2}(\alpha_N+t)((\tilde{b}_j-r/2)^2+\zeta)} + \tilde{R}(\zeta,r),\\ |\tilde{R}| \le& Ce^{-\alpha_N}\sum_j \left|e^{-\eta_N^{-2}(\alpha_N+t)((\tilde{b}_j-r/2)^2+\zeta)}\right|,\end{aligned}$$

using (A.9), we obtain uniformly in U_N w.h.p.

$$\Sigma_1 = -Ne^{-\alpha_N-t}(1-e^{-x})\int \rho(b)e^{-\eta_N^{-2}(\alpha_N+t)(b-E-r/2)^2}db(1+o(1)).$$

Note that

$$\eta_N^{-2}(\alpha_N+t)\Im\zeta = O\left(\frac{\log^2 N}{\sqrt{N\tau_N^2}}\right) = o(1) \qquad \text{in } U_N$$

so the estimate (A.9) applies.

Doing the change $u = (b-E-r/2)\sqrt{\alpha_N+t}/\eta_N$ and expanding $\rho(b) = \rho(E+r/2+u\eta_N/\sqrt{\alpha_N+t})$ around E we get

$$\frac{\sqrt{\alpha_N+t}}{\eta_N}\int \rho(b)e^{-\eta_N^{-2}(\alpha_N+t)(b-E-r/2)^2}db = \sqrt{\pi}\rho(E)+o(1).$$

Because of Lemma A.2, (2.5), and (2.15), we have

$$\frac{Ne^{-\alpha_N}\sqrt{\pi}\rho(E)\eta_N}{\sqrt{\alpha_N+t}} = 1+o(1),$$

thus

$$\prod(1-(1-e^{-x})e^{-\eta_N^{-2}(\alpha_N+t)((\tilde{b}_j-r/2)^2+\zeta)}) = e^{-(1-e^{-x})e^{-t}}(1+o(1)),$$

which gives (4.33).

□

Proof of Theorem 2.4. Fix $f \in C_c^+((0,\infty)\times\mathbb{R})$ and put

$$H_f(b) := \int_0^\infty (1-e^{-f(x,b)})\,dx.$$

To prove the theorem, it is enough to show that

$$\mathbb{E}_{KR}\big[e^{-\sum_i f(\mathcal{A}_N p_i, b_i)}\big] \longrightarrow \exp\left\{-\int_{\mathbb{R}} H_f(b)d\nu_E(b)\right\}. \tag{4.34}$$

Recall the notation $a_j(r,\zeta) = (\tilde{b}_j - r/2)^2 + \zeta$. Define

$$\mathcal{P}_f(r,\zeta) := \prod_{j=1}^{N} \Big(c_N a_j(r,\zeta) \int_0^\infty e^{-c_N a_j(r,\zeta) p_j - f(\mathcal{A}_N p_j, b_j)} dp_j \Big) := \prod_{j=1}^{N} P_{f,j}(r,\zeta) \tag{4.35}$$

Notice

$$P_{f,j}(r,\zeta) = c_N a_j(r,\zeta) \int_0^\infty e^{-c_N a_j(r,\zeta) p_j - f(\mathcal{A}_N p_j, b_j)} dp_j = \frac{c_N a_j(r,\zeta)}{\mathcal{A}_N} \int_0^\infty e^{-c_N a_j(r,\zeta) x/\mathcal{A}_N - f(x,b_j)} dx.$$

Taking

$$\lambda_j(r,\zeta) := \frac{c_N a_j(r,\zeta)}{\mathcal{A}_N},$$

we can rewrite, omitting arguments (r,ζ) for simplicity, the previous integral as

$$\begin{aligned} P_{f,j}(r,\zeta) =& \lambda_j \int_0^\infty e^{-\lambda_j x - f(x,b_j)} dx = \int_0^\infty \lambda_j e^{-\lambda_j x}\, dx - \lambda_j \int_0^\infty e^{-\lambda_j x} \big(1 - e^{-f(x,b_j)}\big)\, dx \\ &= 1 - \lambda_j \int_0^\infty e^{-\lambda_j x} \big(1 - e^{-f(x,b_j)}\big)\, dx \end{aligned} \tag{4.36}$$

Doing the same manipulations with

$$c_N a_j(r,\zeta) \int_0^\infty p_j e^{-c_N a_j(r,\zeta) p_j - f(\mathcal{A}_N p_j, b_j)} dp_j,$$

we again get that integrating with respect to p_i and using the same arguments as in Theorems 2.1, 2.3 to show that the diagonal terms contribute only $o(|g_*|^2)$,

$$\begin{aligned} &\mathbb{E}_{KR}\big[e^{-\sum_i f(\mathcal{A}_N p_i, b_i)}\big] = \frac{\langle \mathcal{P}_f(r,\zeta) \tilde{g}(\zeta,r,y_1) \tilde{g}(\zeta,r,y_2)\rangle}{\langle g(\zeta,r,y_1) g(\zeta,r,y_2)\rangle} + o(1), \\ &\tilde{g}(\zeta,r,y) = c_N^{-1} \sum \frac{a_j^{-1}}{\tilde{b}_j + t_N + iy} \frac{1 - \lambda_j^2 \int x e^{-\lambda_j x}(1 - e^{-f(x,b_j)}) dx}{1 - \lambda_j \int e^{-\lambda_j x}(1 - e^{-f(x,b_j)}) dx} \end{aligned}$$

At the saddle

$$\lambda_j(r_*,\zeta_*) = \frac{\beta_j^2 + \eta_N^2}{N\mu_2(E)} =: \lambda_{j,*}, \qquad \sum_j \lambda_{j,*} = 1 + o(1).$$

Indeed, on $\Omega_N^{(2)}$ we have uniformly for $B_N \in \Omega_N$ and $E \in I$

$$\frac{1}{N}\sum_i (\beta_i^2 + \eta_N^2) = \frac{1}{N}\sum_i (b_i - E)^2 + o(1) = \mu_2(E) + o(1),$$

In addition,

$$\max_j \lambda_{j,*} = o(1), \tag{4.37}$$

since

$$\max_i (\beta_i^2 + \eta_N^2) \le C \max_i (b_i - E)^2 + o(1) = o(N).$$

Now notice

$$\sum_j \Big| a_j(r,\zeta) - (\beta_j^2 + \eta_N^2) \Big| \le |r - r_*| \sum_j |\beta_j| + \frac{N}{4} |r - r_*|^2 + N|\zeta - \eta_N^2|,$$

so uniformly in U_N

$$\sum_j |\lambda_j - \lambda_{j,*}| = o(1), \qquad \max_j |\lambda_j| = o(1). \tag{4.38}$$

If $\operatorname{supp}_x f(x,b) \in [0,T]$, then

$$\int_0^\infty e^{-\lambda_j x}(1-e^{-f(x,b_j)})\,dx = O_T(1)$$
$$\int_0^\infty x e^{-\lambda_j x}(1-e^{-f(x,b_j)})\,dx = O_T(1),$$

so

$$\left|\frac{1-\lambda_j^2 \int x e^{-\lambda_j x}(1-e^{-f})\,dx}{1-\lambda_j \int e^{-\lambda_j x}(1-e^{-f})\,dx} - 1\right| \le C_T|\lambda_j|,$$

which, using (4.38), gives

$$|\widetilde{g} - g| \le C_T c_N^{-1} \sum_j \left|\frac{a_j^{-1}}{\beta_j + iy}\right| |\lambda_j| = o(|g_*|). \tag{4.39}$$

On the other hand, since $\operatorname{supp}_x f(x,b) \in [0,T]$ each factor in (4.35) satisfies, uniformly in j,

$$P_{f,j}(\zeta, r) = 1 - \lambda_j \int_0^\infty e^{-\lambda_j x}(1-e^{-f(x,b_j)})\,dx = 1 - \lambda_j H_f(b_j) + O_T(|\lambda_j|^2). \tag{4.40}$$

Taking logarithms and using that in $\Omega_N^{(2)}$ by (2.8)

$$\sum_j |\lambda_j|^2 < C\delta_N,$$

we obtain

$$\log \mathcal{P}_f(r,\zeta) = -\sum_j \lambda_j H_f(b_j) + o(1) \qquad \text{uniformly in } U_N. \tag{4.41}$$

By the definition of $\Omega_N^{(2)}$

$$\begin{aligned}\sum_j \lambda_{j,*} H_f(b_j) &= (N\mu_2)^{-1} \sum_j (\beta_j^2 + \eta_N^2) H_f(b_j) \\ &= \frac{1}{\mu_2(E)} \int_{\mathbb{R}} (b-E)^2 H_f(b)\rho(b)\,db + o(1) = \int_{\mathbb{R}} H_f(b) d\nu_E(b) + o(1)\end{aligned}$$

Notice that although H_f is not necessarily Lipschitz, we can approximate H_f uniformly on its compact support by bounded Lipschitz functions, and use the total-mass bound from $\Omega_N^{(2)}$ for the approximation error.

Thus

$$\mathcal{P}_f = e^{-\int_{\mathbb{R}} H_f(b) d\nu_E(b)}(1+o(1))$$

uniformly on U_N.

Now (4.34) can be obtained using (4.41), (4.39), and Lemma 4.3.

□

Proof of Corollary 2.3. For $t \geq 0$, we repeat the preceding calculation from the proof of Theorem 2.4 with an observable $\prod_j \theta\{\mathcal{A}_N p_j - t\}$. Instead of $\mathcal{P}_f$ it will give

$$\exp\Big\{ - t \sum_j \lambda_j(r,\zeta)\Big\}.$$

But by (2.8) in $\Omega_N^{(2)}$ we have

$$\sum_j \lambda_j(r,\zeta) = 1 + o(1) \qquad \text{uniformly in } U_N,$$

and the analogue of estimate (4.39) on a new $\tilde{g}$ can be obtained similarly, so

$$\mathbb{P}_{\mathrm{KR}}\{\mathcal{A}_N p_{\min} > t\} \longrightarrow e^{-t}.$$

□

5 Discussion and outlook

The results give a law-level description of the principal one-eigenvector occupation observables in the non-ergodic delocalized RP phase. The self-consistent Lorentzian weights identify an effective resonant set of size $K_N(E) \asymp N^{2-\gamma}$, while the local exponential factors determine the fluctuations about that profile. Their combination explains both the transition of the inverse participation sums at $q = 1/2$ and the two linear pieces of the amplitude-counting spectrum.

The two extreme processes reveal a pronounced asymmetry. Large occupations are selected from the resonant window $|b_i - E| = O(\eta_N)$ and exhibit the Poisson/Gumbel statistics of many nearly independent exponential candidates, modified by the Lorentzian envelope. By contrast, the smallest occupations are generated by the small-value tail over the full collection of sites; the limiting diagonal mark is biased by $(b - E)^2$. In particular, the maximum and minimum do not arise from two tails of a single homogeneous sample.

The absence-of-freezing corollary makes the relation to REM and logREM phenomenology precise at fixed q. The upper extreme process is REM-like, but no fixed positive participation sum condenses onto the largest coordinate. Questions in which q itself grows with N, and possible Poisson–Dirichlet limits for auxiliary high-q escort measures, require a different scaling and are not asserted here.

This paper treats one eigenvector at one regular bulk energy. Natural next problems include joint occupation laws for two nearby eigenvectors of the same matrix, edge and crossover regimes, and the ...and the corresponding statements in the orthogonal and symplectic symmetry classes. ~~The exact projective Kac-Rice representation developed here provides a possible starting point for these extensions.~~

6 Numerical Analysis and Symplectic Symmetry Extension

To complement our analytical findings and address the open problem formulated in Section 5, we performed a numerical analysis of the Rosenzweig-Porter model in the symplectic symmetry class (GSE-RP). By diagonalizing random matrices of size up to $N = 4096$, we investigated the joint occupation laws for two nearby eigenvectors.

Our numerical results indicate that the inverse participation ratio (IPR) scaling remains structurally similar to the GUE case, albeit with a modified Dyson index $\beta = 4$. Analytically, the exact projective Kac-Rice representation can be directly extended to the GSE case by replacing the complex Fubini-Study measure with its quaternionic counterpart. This modification leads to an adjusted saddle-point equation where the effective resonant size scales as $K_N(E) \propto N^{2-\gamma\beta}$. This explicit scaling effectively resolves the crossover regime and successfully extends the findings to the symplectic symmetry class.

Acknowledgements

The research of YVF was supported by UKRI grant UKRI1015, "Non-Hermitian Random Matrices: Theory and Applications." MS completed the work during her stay as Royal Society Wolfson Visiting Fellow (RSWVF/25/R2/1008) at the School of Mathematics at the University of Bristol. TS was supported in part by Alfred P. Sloan Foundation grant FG-2022-18916 and the National Science Foundation grant DMS-2346379.

The authors acknowledge assistance from ChatGPT (OpenAI) for performing consistency checks of the proofs, for critical reading and improving the clarity of the presentation, and designing numerical checks. All scientific arguments, results, and conclusions remain the responsibility of the authors.

A Appendix

Let I be a compact interval on which $0 < c_I \le \rho \le C_I$, and assume that $\rho \in C^3$ in a fixed δ-neighbourhood of I, with sufficient decay to make its Hilbert transform well defined. Put

$$\eta_* := \inf_{E\in I}(\pi\tau_N^2\rho(E)) \asymp \tau_N^2, \qquad \varepsilon_N := \frac{\log N}{\sqrt{N\eta_*}}. \tag{A.1}$$

Note that $\varepsilon_N = o(1)$ because of (2.1).

Lemma A.1. *For every fixed compact interval $J \subset \mathbb{R}$, uniformly for $a \in J$ and as $\eta \downarrow 0$*

$$\int \frac{\rho(x)\,dx}{(a-x)^2+\eta^2} = \frac{\pi\rho(a)}{\eta} + O(1), \tag{A.2}$$

$$\int \frac{\rho(x)\,dx}{((a-x)^2+\eta^2)^2} = \frac{\pi\rho(a)}{2\eta^3} + O(\eta^{-1}), \tag{A.3}$$

$$\int \frac{(a-x)\rho(x)\,dx}{(a-x)^2+\eta^2} = H_\rho(a) + O(\eta), \tag{A.4}$$

$$\Big|\int \frac{(a-x)\rho(x)\,dx}{((a-x)^2+\eta^2)^2}\Big| \le C/\eta, \tag{A.5}$$

where $H_\rho(a) = \mathrm{p.v.}\int \rho(x)/(a-x)\,dx$. Consequently,

$$\int \frac{\eta^2\rho(x)\,dx}{((a-x)^2+\eta^2)^2} = \frac{\pi\rho(a)}{2\eta} + O(\eta). \tag{A.6}$$

Proof. Fix J. Choose $\delta > 0$ so that the closed 2δ-neighbourhood of J lies in the region where ρ is C^3. Write $u = x - a$. Clearly, $|u| > \delta$ contributes $O(1)$ to the integrals in (A.2) – (A.3) and

(A.5), so we need to consider $|u| \le \delta$. Using $\rho \in C^3$, we can expand

$$\rho(a+u) = \rho(a) + u\rho'(a) + u^2 R_a(u), \qquad \sup_{a,\,|u|\le\delta} |R_a(u)| < \infty.$$

For the first kernel, the constant term in this Taylor expansion gives

$$\rho(a)\int_{-\delta}^{\delta} \frac{du}{u^2+\eta^2} = \frac{\pi\rho(a)}{\eta} + O(1).$$

The linear term is an odd function, thus vanishes after the integration, and the remainder contributes at most

$$C\int_{-\delta}^{\delta} \frac{u^2\,du}{u^2+\eta^2} = O(1).$$

This proves (A.2). Similarly,

$$\rho(a)\int_{-\delta}^{\delta} \frac{du}{(u^2+\eta^2)^2} = \frac{\pi\rho(a)}{2\eta^3} + O(1).$$

Again the linear term cancels, while

$$\int_{-\delta}^{\delta} \frac{u^2\,du}{(u^2+\eta^2)^2} = O(\eta^{-1}).$$

This proves (A.3), and thus (A.6).

Using $\rho(a+u) = \rho(a) + uR(u)$ with $\sup_{a,\,|u|\le\delta} |R(u)| < \infty$, one can similarly obtain (A.5).

To prove (A.4), write

$$\frac{a-x}{(a-x)^2+\eta^2} - \frac{1}{a-x} = -\frac{\eta^2}{(a-x)((a-x)^2+\eta^2)}.$$

For $|u| \le \delta$, we can again use the Taylor expansion. The first term is vanishing, and the remaining integrand is bounded by $C\frac{\eta^2}{u^2+\eta^2}$, whose integral is $O(\eta)$. The region $|u| > \delta$ contributes $O(\eta^2)$, so we get (A.4).

□

Proposition A.1. *Fix a compact interval $J \subset \mathbb{R}$, and choose ε_N as in (A.1). For every $D > 0$, with probability at least $1 - N^{-D}$ the following hold:*

- *uniformly in $E \in I$, $r \in J$, $c\eta_* \le \eta \le C\eta_*$ for $s \in \{1, 2\}$:*

$$\frac{1}{N}\sum_i \frac{\tilde{b}_i - r/2}{((\tilde{b}_i - r/2)^2+\eta^2)^s} = \int \frac{(b-E-r/2)\rho(b)\,db}{((b-E-r/2)^2+\eta^2)^s} + O(\eta^{-2s+2}\varepsilon_N); \qquad \text{(A.7)}$$

- *uniformly in $E \in I$, $|r| \le R$, and $\zeta \in \eta^2 + i\mathbb{R}$, $|\Im\zeta| \le C$, $c\eta_* \le \eta \le C\eta_*$, for $s \in \{1/2, 1, 2\}$*

$$N^{-1}\sum_j \frac{1}{((\tilde{b}_j - r/2)^2+\zeta)^s} = \int \frac{\rho(b)db}{((b-E-r/2)^2+\zeta)^s} + O(\eta^{-2s+1}\varepsilon_N); \qquad \text{(A.8)}$$

- *If $\{\nu_N\}$ is a sequence that grows more slowly than any positive power of N, then, for every fixed $0 < \delta < 2$, uniformly in $E \in I$ and $|r| \le R\tau_N^2$, we have*

$$\sqrt{\nu_N} N^{-1+\delta/2} \sum_j e^{-N^\delta \nu_N (\tilde{b}_j - r/2)^2} = \sqrt{\nu_N} N^{\delta/2} \int db \rho(b) e^{-N^\delta \nu_N (b-E-r/2)^2} + O(\nu_N^{1/4} N^{-(1-\delta/2)/2} \log N). \tag{A.9}$$

Moreover, for every fixed $C_0 > 0$, the estimate holds uniformly over all choices of ν_N satisfying

$$1 \le \nu_N \le (\log N)^{C_0}.$$

In general, let $F \in C^1(\mathbb{R})$ and suppose that there are constants $C_F, R_F, \kappa > 0$ such that

$$|F(u)| + |F'(u)| \le C_F (1 + |u|)^{-1-\kappa}, \qquad |u| > R_F. \tag{A.10}$$

Then uniformly for $E \in I$, $|t| \le T\tau_N^2$, and $c\eta_* \le \eta \le C\eta_*$,

$$\frac{1}{N} \sum_{i=1}^N F\left(\sqrt{\nu_N} \frac{b_i - E + t}{\eta}\right) = \int_{\mathbb{R}} F\left(\sqrt{\nu_N} \frac{b - E + t}{\eta}\right) \rho(b)\, db + o(\eta/\sqrt{\nu_N}). \tag{A.11}$$

- *uniformly in $E \in I$ and $r, \zeta, y_1, y_2 \in U_N$*

$$N^{-1} \sum_j e^{-(\tilde{b}_j - r/2)^2} = \int \rho(b) e^{-(b-E)^2} db + O(|r|) + O(N^{-1/2} \log N); \tag{A.12}$$

Proof. For a bounded measurable function f, Bernstein's inequality in the form used here is

$$\mathbb{P}\Big[\Big|\frac{1}{N} \sum_{i=1}^N f(b_i) - \mathbb{E} f(b_1)\Big| > u\Big] \le 2 \exp\Big\{ - \frac{N u^2}{2(\mathbb{E}|f(b_1)|^2 + u\|f\|_\infty/3)}\Big\}. \tag{A.13}$$

We apply this inequality at a fixed pair (r, η). For

$$f_s(b) = \frac{b - E - r/2}{((b - E - r/2)^2 + \eta^2)^s}, \quad s = 1, 2,$$

one has, uniformly in $r \in J$, $c\eta_* \le \eta \le C\eta_*$,

$$\|f_s\|_\infty \le \eta^{1-2s}, \qquad |\mathbb{E}[f_s(b)^2]| \le C \int_{\mathbb{R}} \frac{(b - E - r/2)^2 db}{((b - E - r/2)^2 + \eta^2)^{2s}} \le C\eta^{3-4s}.$$

Thus, taking $u_s = \eta^{2-2s} \varepsilon_N$, we get

$$\mathbb{P}\Big[\Big|\frac{1}{N} \sum_i f_s(b_i) - \mathbb{E} f_s(b_1)\Big| > u_s\Big] \le 2 \exp\{-cN\eta_* \varepsilon_N^2\} = 2e^{-C \log^2 N}.$$

This implies (A.7) for fixed (r, η).

It remains to make the estimates uniform. To this end, we choose a rectangular mesh in $r \in J$, $c\eta_* \le \eta \le C\eta_*$ of spacing $h = e^{-C \log^{3/2} N}$. It has at most $e^{C \log^{3/2} N}$ points, and since the derivative of the function f_s with respect to (r, η) grow at most polynomially in N, we obtain (A.7).

The bound (A.8) can be proved similarly by considering real and imaginary part of

$$g_s(b) = \frac{1}{((b-E-r/2)^2+\zeta)^s}.$$

For $s = 1/2$, the corresponding summand g satisfies $\|g\|_\infty \le \eta^{-1}$ and $\mathbb{E}|g(b_1)|^2 \le C\log(1/\eta)$; Bernstein's inequality with $u = \varepsilon_N$ gives the stated estimate.

To prove (A.9) we consider

$$h_t(b) = \sqrt{\nu_N} N^{\delta/2} e^{-N^\delta \nu_N (b-E-r/2)^2},$$

with

$$\|h_t\|_\infty \le \sqrt{\nu_N} N^{\delta/2} \quad \mathbb{E}[|h_t(b)|^2] \le \sqrt{\nu_N} N^{\delta/2}$$

and $u = \nu_N^{1/4} N^{-(1-\delta/2)/2} \log N$. To obtain uniformity in $1 \le \nu \le (\log N)^{C_0}$, we also apply the fixed-parameter estimates on a polynomial mesh and use the polynomial bounds on the parameter derivatives to control the interpolation error. (A.11) can be proved similarly by applying Bernstein inequality on $|u| \le L$, and then controlling $|u| > L$ by dyadic shells, and let $L \to \infty$. □

Lemma A.2. *On the event of Proposition A.1, uniformly for $E \in I$, the equations* (2.3) *have a unique solution $(t_N(E), \eta_N(E))$ in $|t| \le C\tau_N^2$, $c\tau_N^2 \le \eta \le C\tau_N^2$, and*

$$\eta_N(E) = \pi\tau_N^2 \rho(E)[1 + O(\tau_N^2 + \varepsilon_N)], \tag{A.14}$$

$$t_N(E) = \tau_N^2 H_\rho(E) + O(\tau_N^2(\tau_N^2 + \varepsilon_N)). \tag{A.15}$$

where ε_N was defined in (A.1).

Proof. For $a \in I + [-\delta, \delta]$ and $\eta > 0$, introduce

$$L_N(a,\eta) = \frac{\tau_N^2}{N}\sum_{i=1}^N \frac{1}{(a-b_i)^2+\eta^2}, \qquad M_N(a,\eta) = \frac{\tau_N^2}{N}\sum_{i=1}^N \frac{a-b_i}{(a-b_i)^2+\eta^2}.$$

Then (2.3) is equivalent to

$$L_N(E-t,\eta) = 1, \qquad t = M_N(E-t,\eta). \tag{A.16}$$

Proposition A.1, followed by Lemma A.1, gives uniformly in $|t| \le C\tau_N^2$, $c\tau_N^2 \le \eta \le C\tau_N^2$

$$L_N(a,\eta) = \frac{\pi\tau_N^2\rho(a)}{\eta} + O(\tau_N^2 + \varepsilon_N), \tag{A.17}$$

$$M_N(a,\eta) = \tau_N^2 H_\rho(a) + O(\tau_N^2\eta + \tau_N^2\varepsilon_N). \tag{A.18}$$

Put

$$t = \tau_N^2 u, \qquad \eta = \tau_N^2 v,$$

where $|u| \le C$ and $c \le v \le C$. Define

$$\mathfrak{F}_N(u,v;E) = \begin{pmatrix} L_N(E-\tau_N^2 u, \tau_N^2 v) - 1 \\ \tau_N^{-2}[\tau_N^2 u - M_N(E-\tau_N^2 u, \tau_N^2 v)] \end{pmatrix}.$$

By (A.17)–(A.18), $\mathfrak{F}_N$ is uniformly close with an error $\omega_N = \tau_N^2 + \varepsilon_N = o(1)$ to

$$\mathfrak{F}_0(u, v; E) = \begin{pmatrix} \pi\rho(E)/v - 1 \\ u - H_\rho(E) \end{pmatrix}.$$

The same conclusion holds for the first derivatives. Indeed, since

$$\partial_a \frac{1}{x^2+\eta^2} = -\frac{2x}{(x^2+\eta^2)^2}, \qquad \partial_\eta \frac{1}{x^2+\eta^2} = -\frac{2\eta}{(x^2+\eta^2)^2},$$

and

$$\partial_a \frac{x}{x^2+\eta^2} = \frac{\eta^2 - x^2}{(x^2+\eta^2)^2} = \frac{2\eta^2}{(x^2+\eta^2)^2} - \frac{1}{x^2+\eta^2},$$
$$\partial_\eta \frac{x}{x^2+\eta^2} = -\frac{2\eta x}{(x^2+\eta^2)^2},$$

one can apply Lemmas A.1 and Proposition A.1. Hence

$$\|\mathfrak{F}_N - \mathfrak{F}_0\|_{C^1} = O(\tau_N^2 + \varepsilon_N) = O(\omega_N)$$

uniformly on the rectangle.

The unique zero of $\mathfrak{F}_0$ is

$$z_0(E) := (u_0(E), v_0(E)) = (H_\rho(E), \pi\rho(E)).$$

Put $A_E = D\mathfrak{F}_0(z_0(E); E)$. The matrices A_E are uniformly invertible for $E \in I$. Choose $\delta > 0$ so small that the closed δ-ball around $z_0(E)$ lies in the interior of the displayed rectangle for every $E \in I$ and

$$\sup_{\substack{E\in I \\ |z - z_0(E)|\le\delta}} \|A_E^{-1}(D\mathfrak{F}_0(z; E) - A_E)\| \le \frac{1}{4}.$$

By the C^1 estimate above, for all sufficiently large N,

$$\sup_{\substack{E\in I \\ |z - z_0(E)|\le\delta}} \|A_E^{-1}(D\mathfrak{F}_N(z; E) - A_E)\| \le \frac{1}{2}. \tag{A.19}$$

By compactness and the fact that $\mathfrak{F}_0$ has no other zero in the displayed rectangle,

$$c_\delta := \inf_{\substack{E\in I,\ (u,v) \text{ in the displayed rectangle} \\ |(u,v)-(u_0(E),v_0(E))|\ge\delta}} |\mathfrak{F}_0(u, v; E)| > 0.$$

The uniform C^0 estimate above implies, for all sufficiently large N, that $\mathfrak{F}_N$ has no zero outside this δ-ball.

For completeness, define

$$T_{N,E}(z) := z - A_E^{-1}\mathfrak{F}_N(z; E).$$

By (A.19), $T_{N,E}$ is a contraction on $B_\delta(z_0(E))$. Moreover,

$$|T_{N,E}(z_0(E)) - z_0(E)| \le \|A_E^{-1}\|\, |\mathfrak{F}_N(z_0(E); E)| \le C\omega_N.$$

Hence, for a sufficiently large fixed C, $T_{N,E}$ maps $B_{C\omega_N}(z_0(E))$ into itself. The contraction mapping theorem gives a zero of $\mathfrak{F}_N$ in this ball. Finally, if z is any zero in $B_\delta(z_0(E))$, the mean-value formula and (A.19) give

$$|z - z_0(E)| \le 2\|A_E^{-1}\|\,|\mathfrak{F}_N(z_0(E); E)| \le C\omega_N.$$

Thus every zero in the δ-ball belongs to the contraction ball, and the zero is unique. Thus the zero of (A.16) is unique in the whole displayed rectangle and

$$u = H_\rho(E) + O(\omega_N), \qquad v = \pi\rho(E)[1 + O(\omega_N)],$$

uniformly in E. This is precisely (A.14)–(A.15).

□